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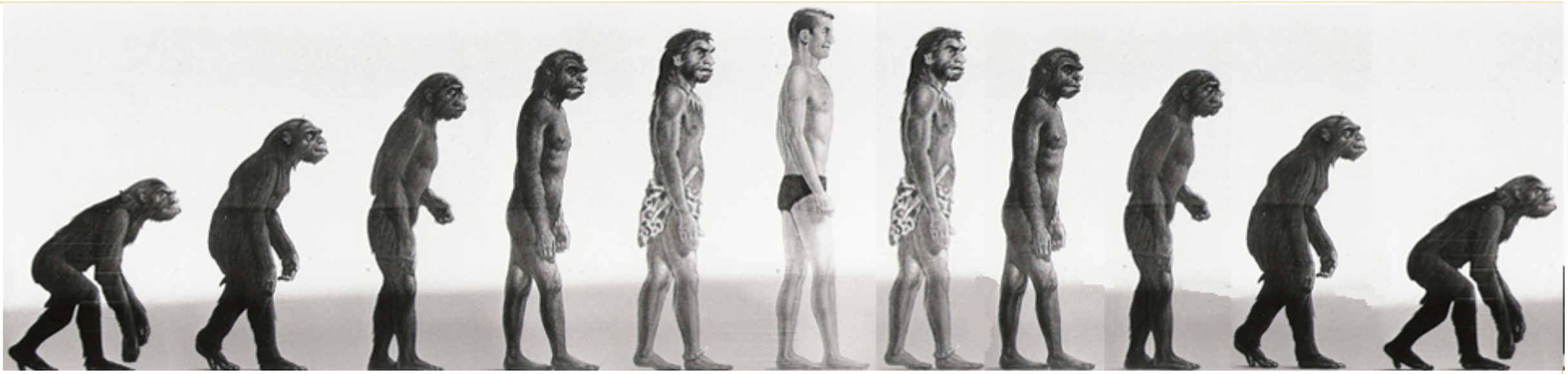

Hominization

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Geo-ecological Background

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Ecological Background

- 70 million years ago, earth was characterized by drastic climatic changes - temperature was falling down and climate was becoming extremely cold.
- Surface of the earth was covered by thick forest and grass
- In such climate, insectivores, small bodied, warm blooded animals proved themselves better adaptive than others
- Since some of these are arboreal, they found less competition and these gave rise to the stock primates.
- As a consequence, primates developed many arboreal evolutionary tendencies

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Geological Changes

- Between 60 and 40 MYBP - various geological forces at work
- Continents were splitting - tectonics
  - South America - Split from Africa
  - North America - Split from Eurasia
  - North and South America - Drifted west wards - Atlantic ocean opened up

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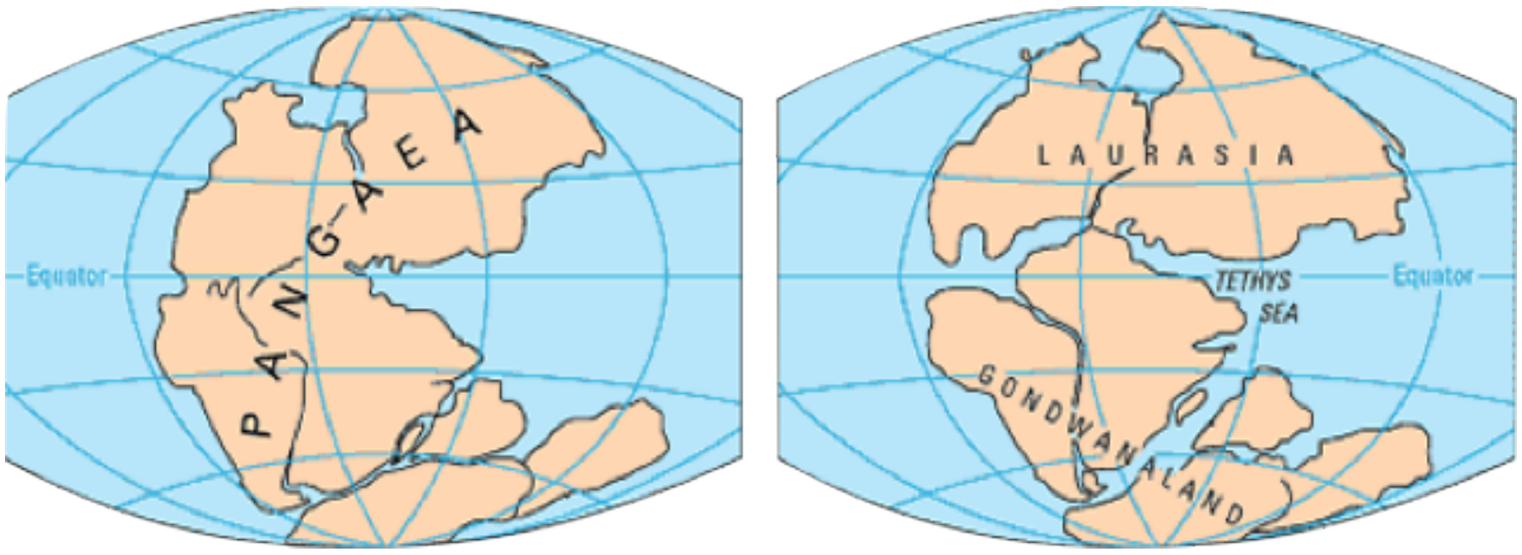
The Plate Tectonics



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Geological Changes

- Before drift - Pangaea contain prosimians
- After drift, both South America and Afro-Asian land mass contained some prosimians
- Prosimians of South America faced a different climate - dense forest - and gave rise to New World Monkeys
- The Old World had less dense forests, open woodlands and no canopy of grass and the prosimians were compelled to drift southwards and they patronized primate evolution - Old World Monkeys - Apes

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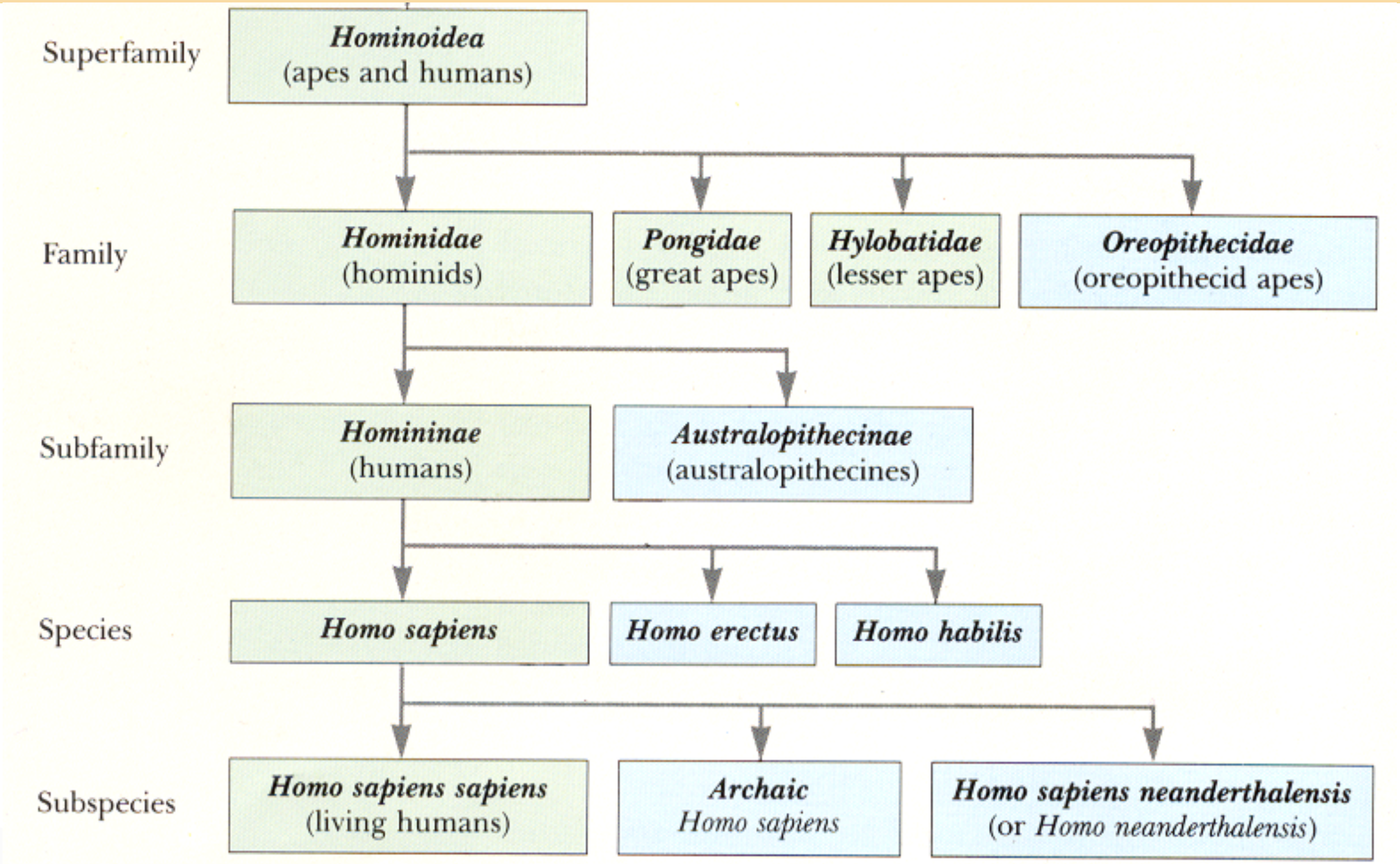
Human phylogeny



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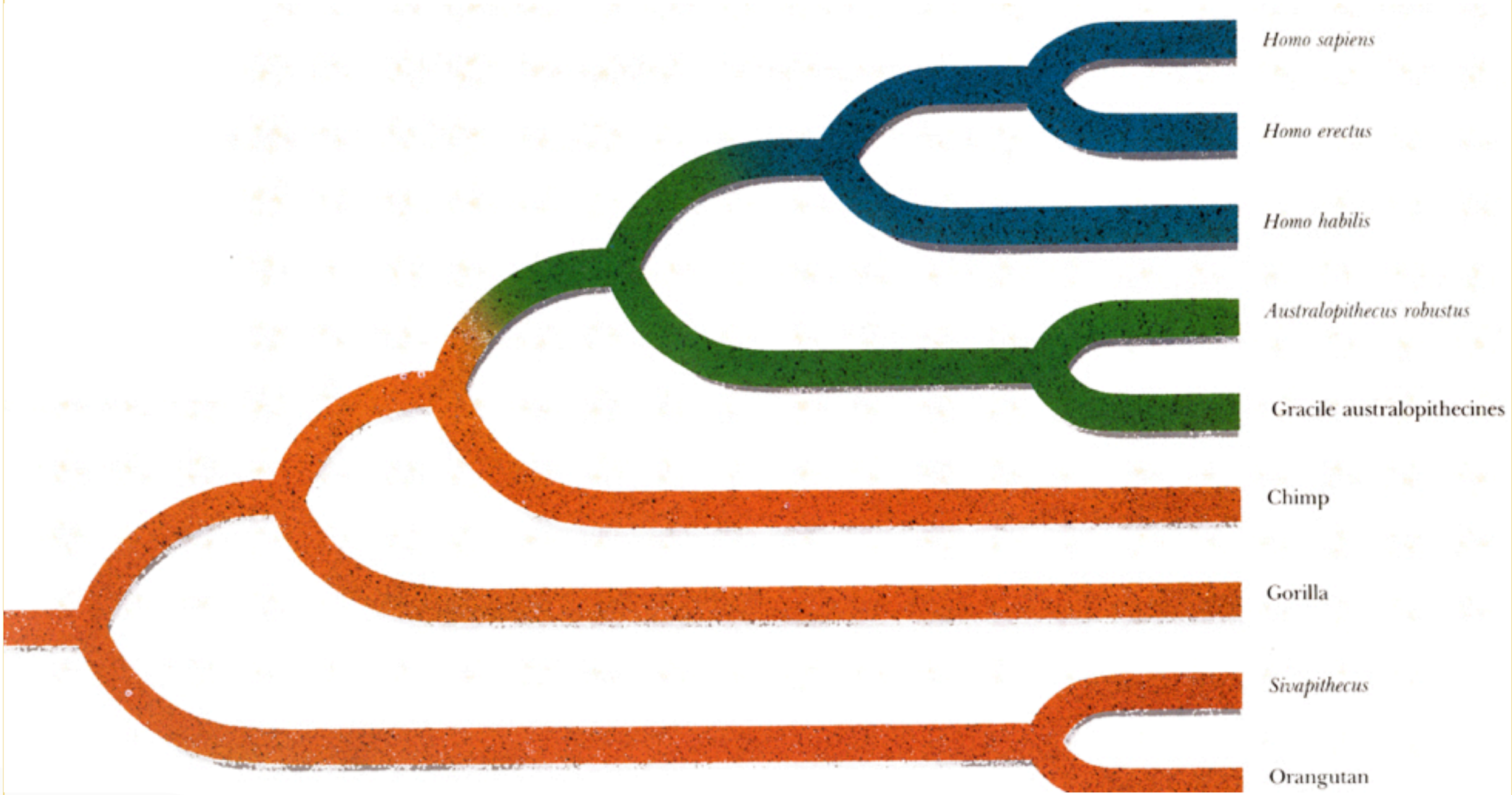
Cladogram (The Evolutionary Tree) of Human Evolution



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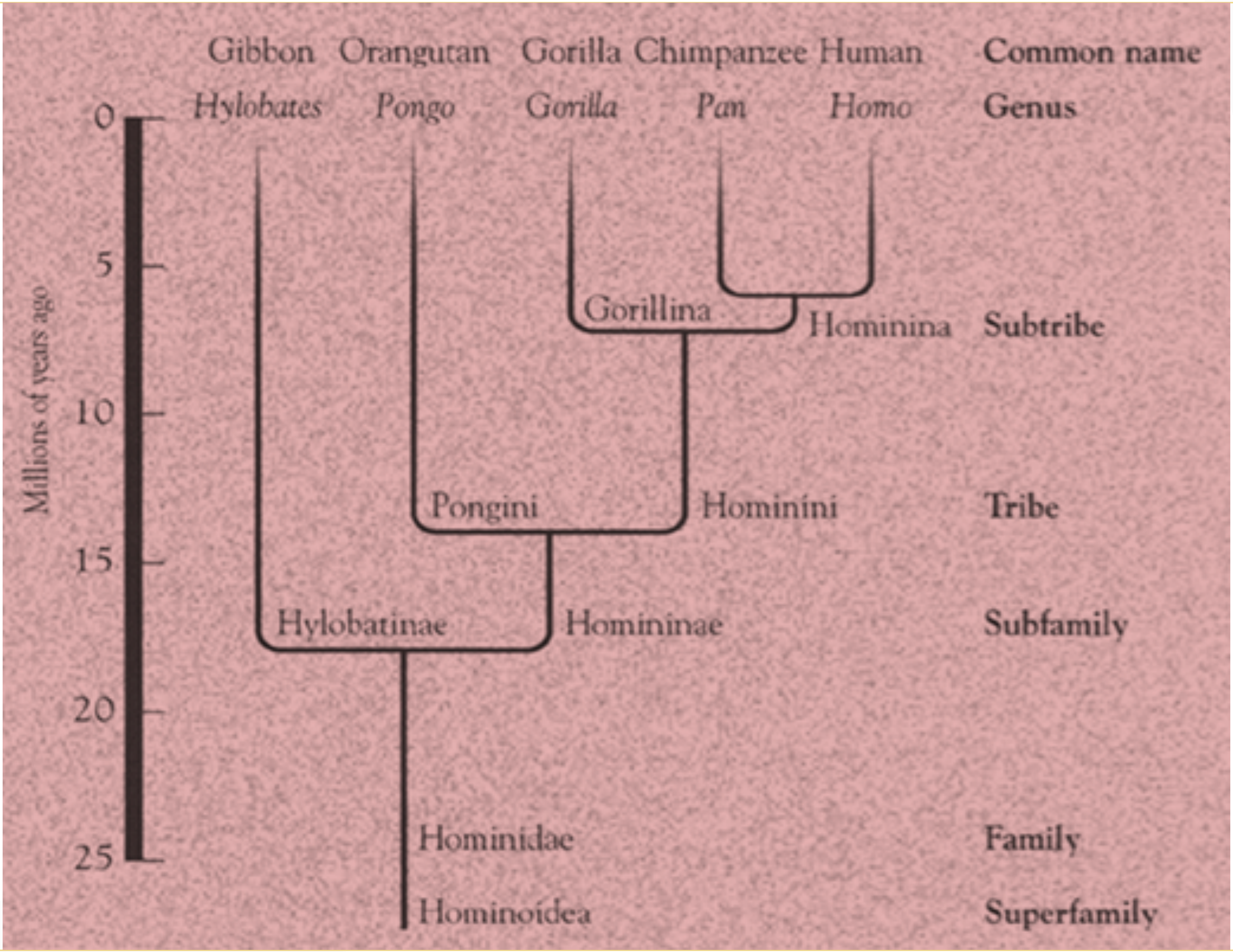

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GEOLOGIC TIME SCALE

Time Units of the Geologic Time Scale					Development of Plants and Animals
Eon	Era	Period		Epoch	
Phanerozoic	Cenozoic	Quaternary		Holocene 0.01	Earliest *Homo sapiens*
				Pleistocene 1.6	
		Tertiary		Pliocene 5.3	Earliest hominids
				Miocene 23.8	
				Oligocene 33.7	"Age of Mammals"
				Eocene 55	
				Palaeocene 65	Extinction of dinosaurs and many other species
	Mesozoic	Cretaceous 145		"Age of Reptiles"	First flowering plants First birds
		Jurassic 208			Dinosaurs dominant First mammals
		Triassic 248			
	Palaeozoic	Permian 286		"Age of Amphibians"	Extinction of trilobites and many other marine animals
		Carboniferous	Pennsylvanian 320		First reptiles Large coal swamps
			Mississippian 360		Amphibians abundant
		Devonian 410		"Age of Fishes"	First amphibians First insect fossils
		Silurian 438			Fishes dominant
		Ordovician 505		"Age of Invertebrates"	First land plants First fishes
		Cambrian 545			Trilobites dominant First organisms with shells
		Vendian 650		"Soft-bodied faunas"	Abundant Ediacaran faunas
Proterozoic 2500		Collectively called Precambrian comprises about 87% of the geological time scale			First multicelled organisms
Archean 3800					First one-celled organisms Age of oldest rocks
Hadean 4600 Ma					Origin of the earth

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GEOLOGIC TIME SCALE

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Eon	Era	Period	Epoch		
Phanerozoic	Cenozoic	Quaternary	Holocene	0.01	Earliest *Homo sapiens*
			Pleistocene	1.6	
		Tertiary	Pliocene	5.3	Earliest hominids
			Miocene	23.8	
			Oligocene	33.7	"Age of Mammals"
			Eocene	55	
			Palaeocene	65	Extinction of dinosaurs and many other species
	Mesozoic	Cretaceous	"Age of Reptiles"	145	First flowering plants First birds
		Jurassic		208	Dinosaurs dominant
		Triassic		248	First mammals
		Permian	"Age of Amphibians"	286	Extinction of trilobites and many other marine animals
		Carboniferous: Pennsylvanian		320	First reptiles Large coal swamps
		Carboniferous: Mississippian			Amphibians abundant

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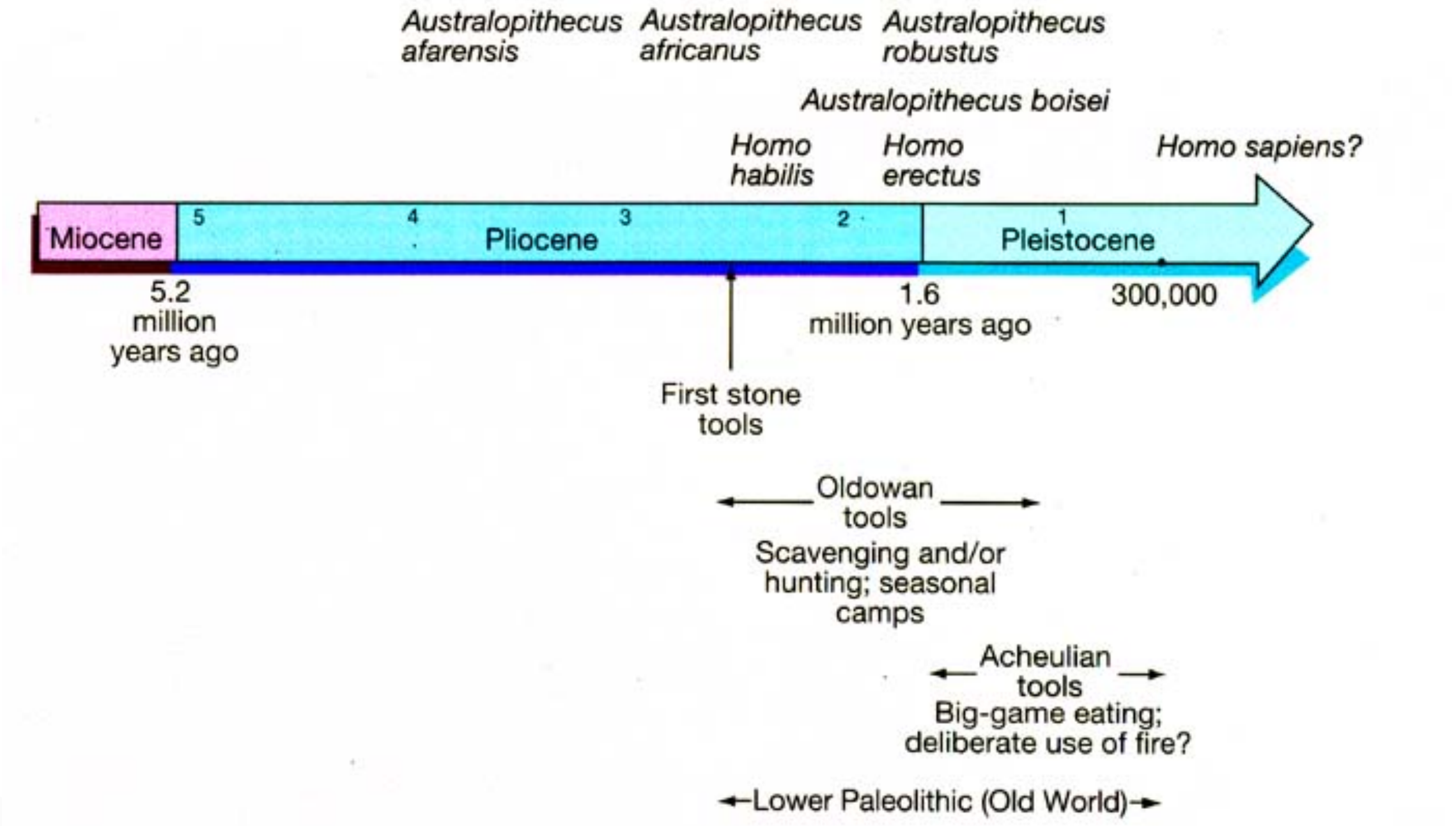
Time of Early Hominids



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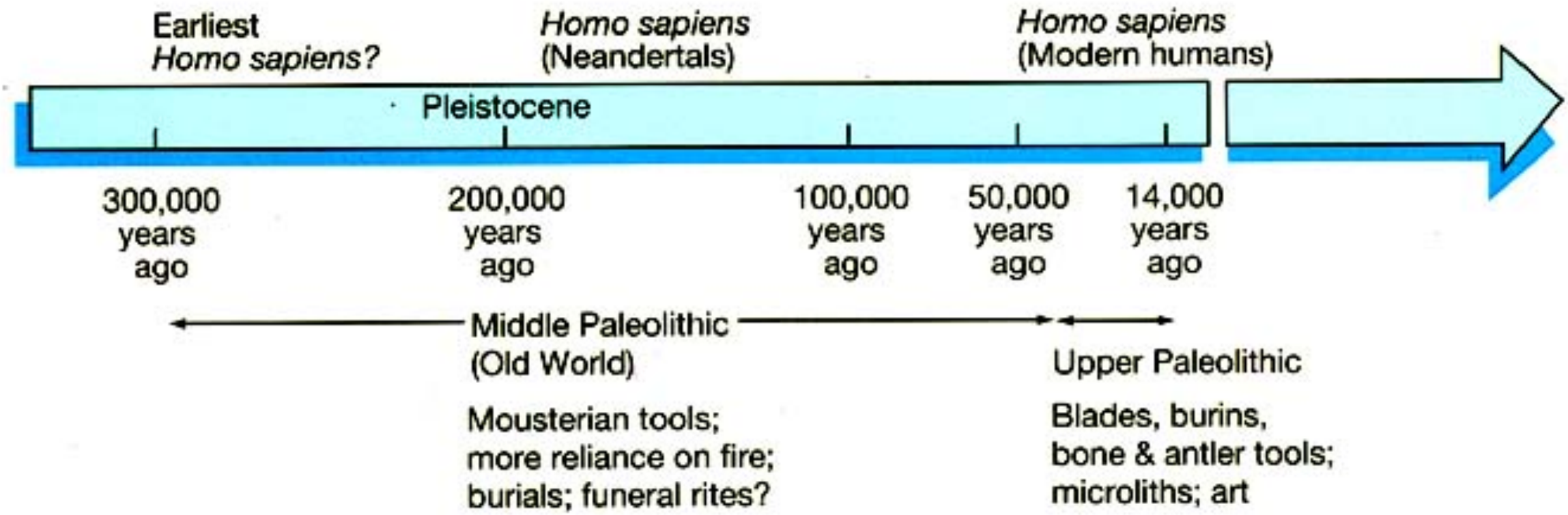
Time of Later Hominids



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The Human Fossils

- Phylogenetic status, characteristics and geographical distribution
  - Plio-pleistocene hominids in South and East Africa - Australopithecines.
  - *Homo erectus: Africa* (*Paranthropus*), Europe (*Homo erectus heidelbergensis*), Asia (*Homo erectus javanicus* , *Homo erectus pekinensis).*
  - Neanderthal Man- La-Chapelle-aux-saints (Classical type), Mt. Carmel (Progressive type).
  - Rhodesian man.
  - *Homo sapiens* -- Cromagnon, Grimaldi and Chancelede.

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Ramapithecus / Sivapithecus

- Last of Miocene apes - 12 MYBP
- First true hominid to embark on the road to human evolution
- Discovery
  - Edward Lewis - Shivalik - Ramapithecus / Sivapithecus
  - Also in Fort Ternan – Kenya (Kenyapithecus)

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Ramapithecus - Climate

- Sub-tropical climate - warm
- Occasional savannas
- No year long fruit period
- No continuous vegetative renewal
- Seasonal forests

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Ramapithecus

- Tendency to find alternate source of food on ground
- Food likely to be seeds, roots and nuts
- This requires tool use / accurate information processing
- This resulted in appearance of hominids with relatively high cultural and intellectual ability

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Characteristics

- Reduced canines and grinding molars (nuts and roots)
- Size - 3½ feet
- Occasional bipedalist - a tendency due to wide field of vision through tall grass
- Small size and small canines - easy prey for savanna carnivores - social grouping? / Semi-erect posture - freed hands - tool use?

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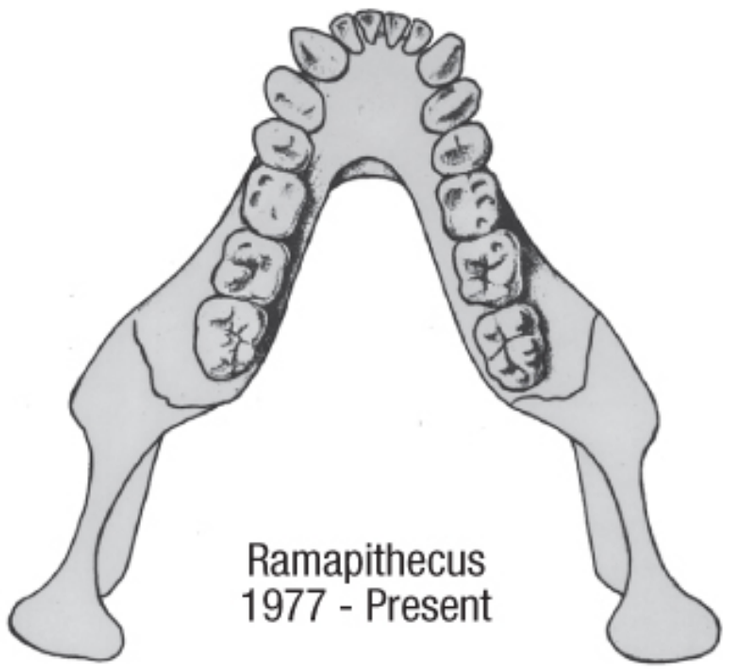
Phylogenetic Implications

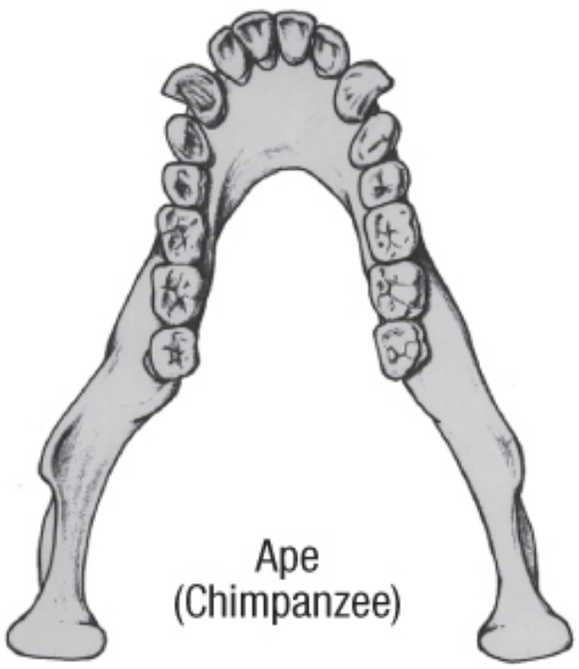
- Not much evidence to exactly justify
- But is a likely candidate for ancestry because in the same areas, in similar environments the next hominids developed
- It might thus have evolved into Australopithecus

Ramapithecus
1932 - 1977

Man



Ramapithecus
1977 - Present



Ape
(Chimpanzee)

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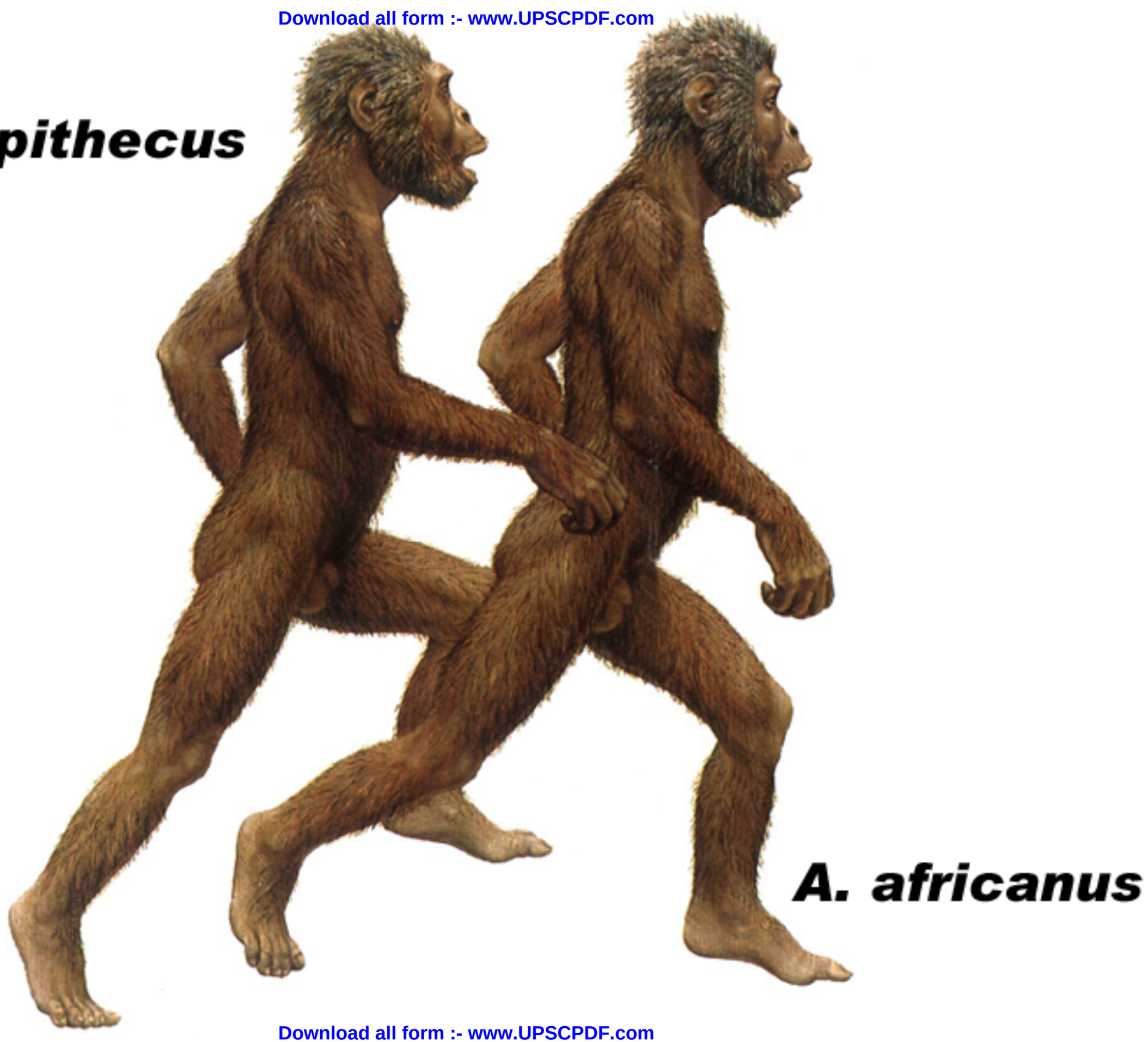

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Australopithecines

- Australopithecines emerged on the rolling savannas of east and south Africa 5-6 MYBP - Pliocene and early Pleistocene
- They show great amount of variation amongst themselves due to localized adaptations
- Paleontologists classified them into two broad groups - **Robust and Gracile**

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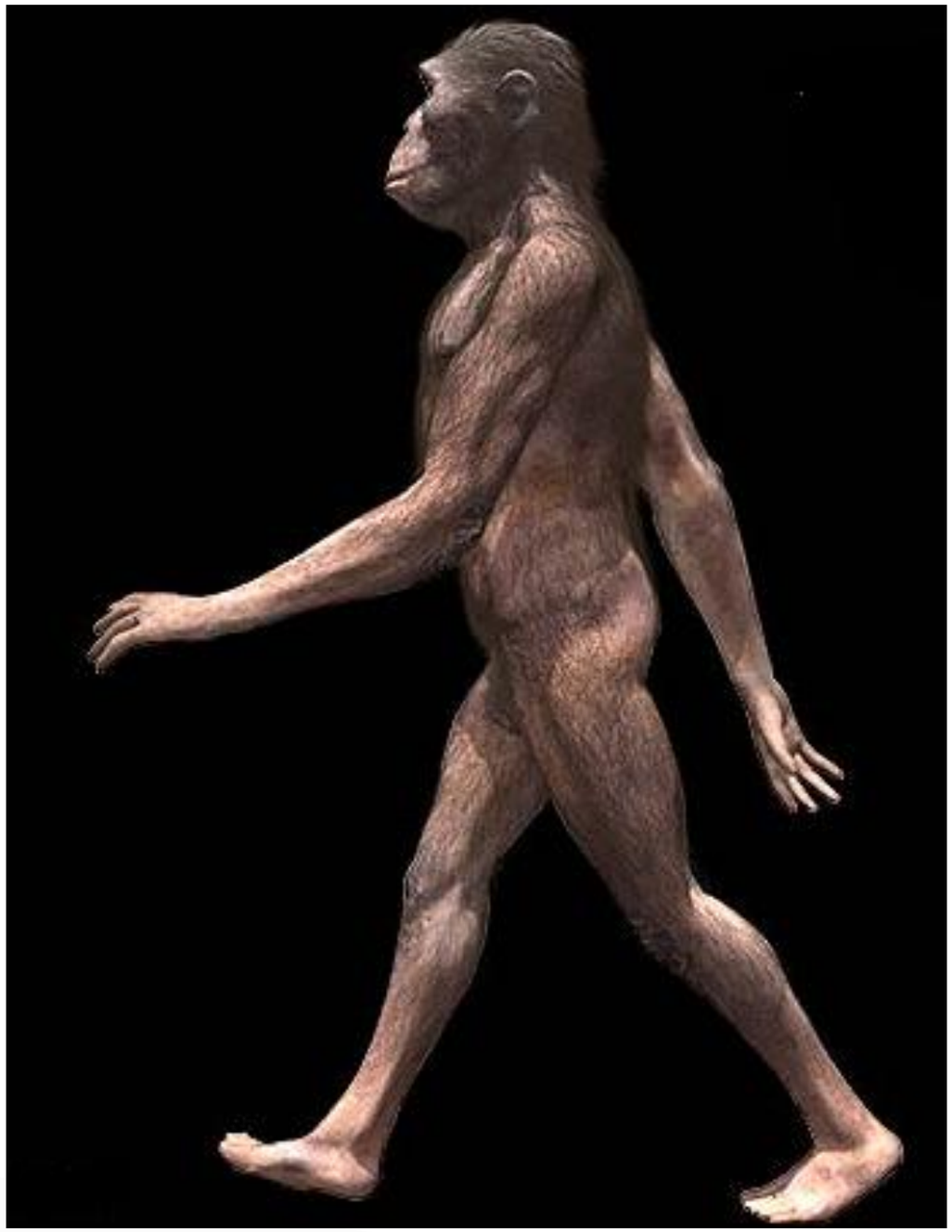
Australopithecus Vs Paranthropus



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Australopithecines

Australopithecus Gracile / Slender	Paranthropus Robust
• A. africanus • A. aferensis • A. ramidus • A. anamensis	• P. robustus • P. bosei

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Discovery

- 1925, Dart discovered the first fossil from Taung in South Africa.
- 1936, Robert Brown discovered an adult cranium from Sterkfontein.
- Dart, Brown and Robinson discovered may more fossils at Kromdraii, Swartkrans, Makapansgat etc.
- 1960s and 1970s many similar fossils - Olduvai George in North Tanzania by Leakey, Hadar in Ethiopia by Johanson and Timothy White.
- 1994 - Richard Leakey from Omo and Koobi Fora
- 1995, Ethiopia by White; Kenya by Leaky.

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Distribution

South **Africa**

- Taung, Cape province - A. africanus;
- Sterkfontein, Transvaal - A. africanus;
- Makapansgat, Transvaal – A. africanus;
- Swartkrans, Transvall - P. robustus.
- Kromdraai, Transvall - P. robustus;

East Africa

- Tanzania - Garusi, Olduvai - P. boisei
- Kenya - Lake Turkana - P. boisei
- Ethiopia - Lake Omo - P. robustus
- Ethiopia - Aramis - A. ramidus
- Kenya - Kanapoi - A. anamensis.

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General Characteristics of Australopithecines

- Roughly 4 feet tall and weighed 50-100 pounds
- No Sagital Crest
- No canine diastema
- Well developed forehead - suggest expansion of frontal lobes of brain
- Bone structure less rugged than their robust relatives

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General Characteristics

- Weakly developed supra orbital ridges
- Elevated cranial vault - indicating major portion of brain above the level of the face unlike pongids where it is behind
- Brain size - 450-600 CC
- Foramen magnum forwardly placed slightly
- Parabolic dental arcade
- Hominid post cranial skeleton

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A. africanus

Smaller Temporal Fossa

Longer Snout

No Crest

Face Sits below Neurocranium

Less Robust Zygomatics

Steeper Forehead

Incisor Prognathism

Less Robust Mandible

Simpler dp_3

Relatively Larger Incisors and Canines

Relatively Smaller Premolars and Molars

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Paranthropus

- Average height - 4 feet and weigh 120 pounds
- Larger skull than Australopithecus
- Cranial capacity - 400-600 CC (lower brain to body size ratio than Australopithecus)
- Large supra orbital ridges (no fore head)

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Paranthropus

- No expanded cranial vault
- Massive jaws
- Parabolic dental arcade - slightly U-shaped
- Post cranial skeleton - mixture of Gorrilloid and hominid traits
- Knuckle walking hand functional anatomy

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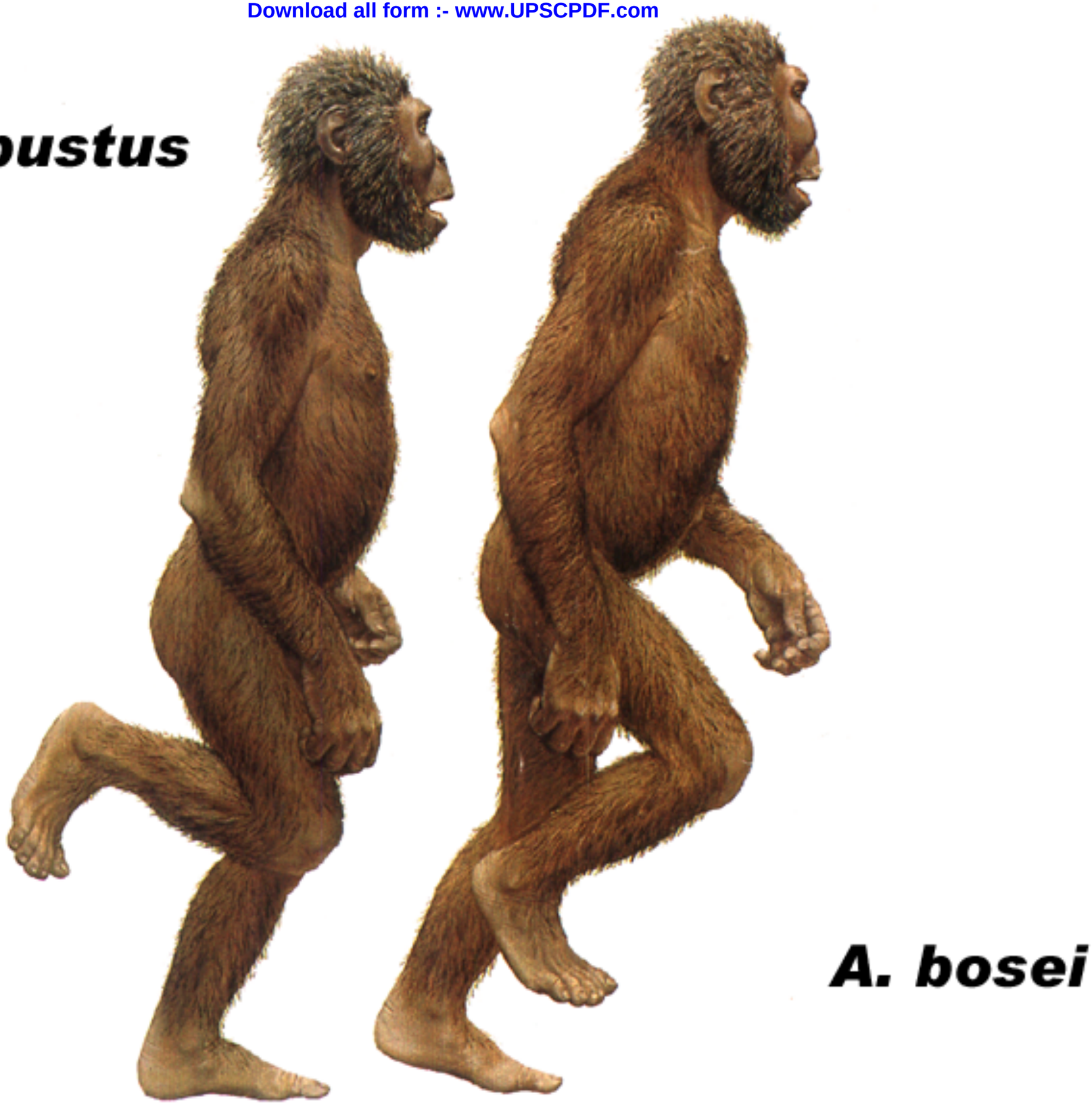

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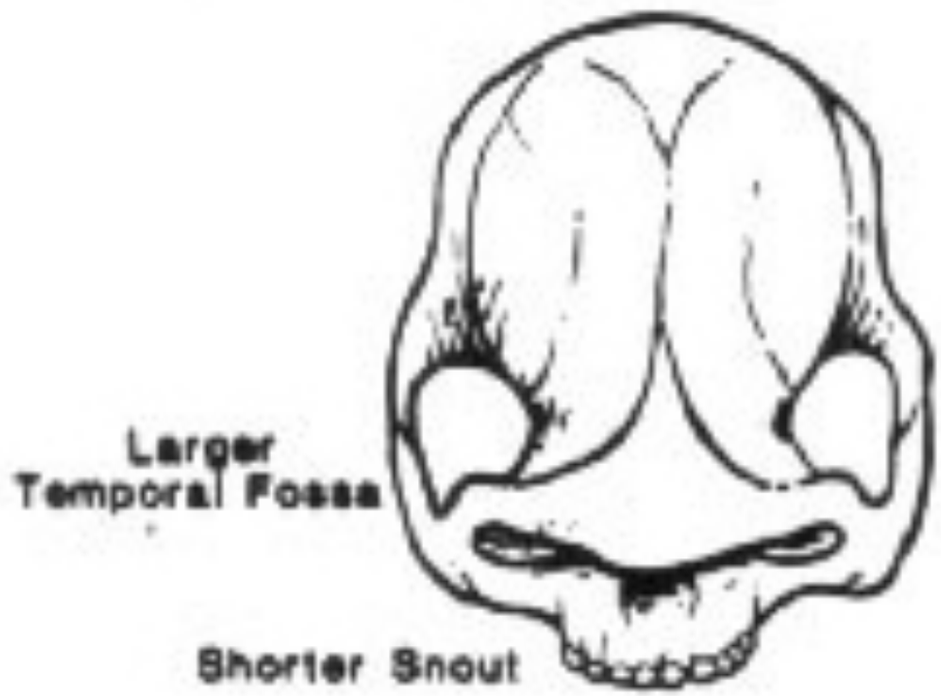
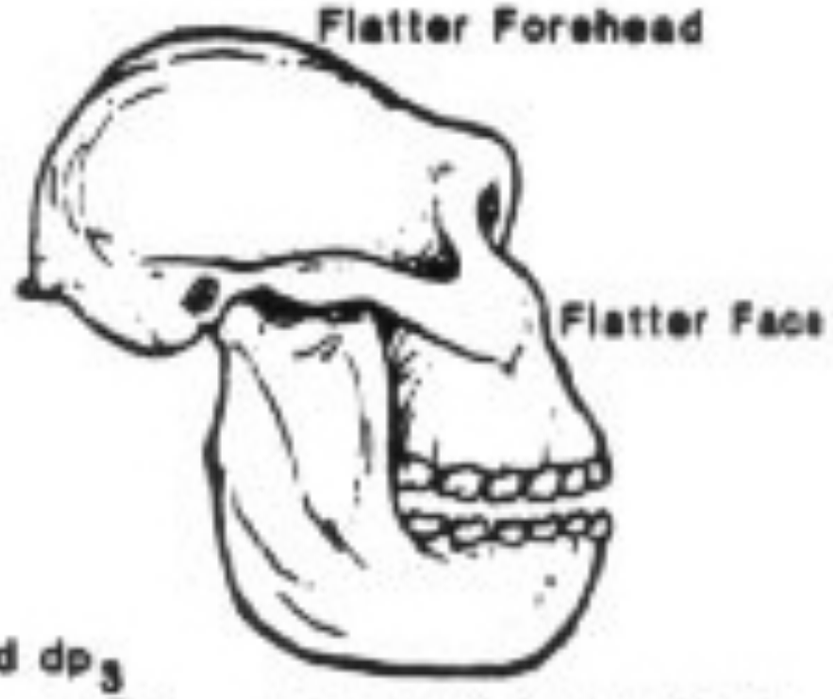
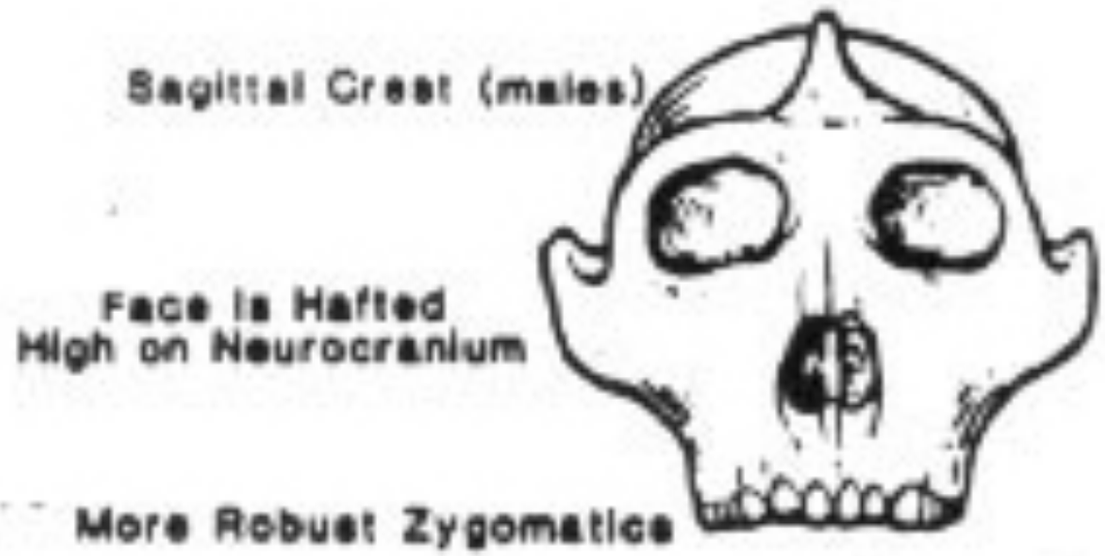
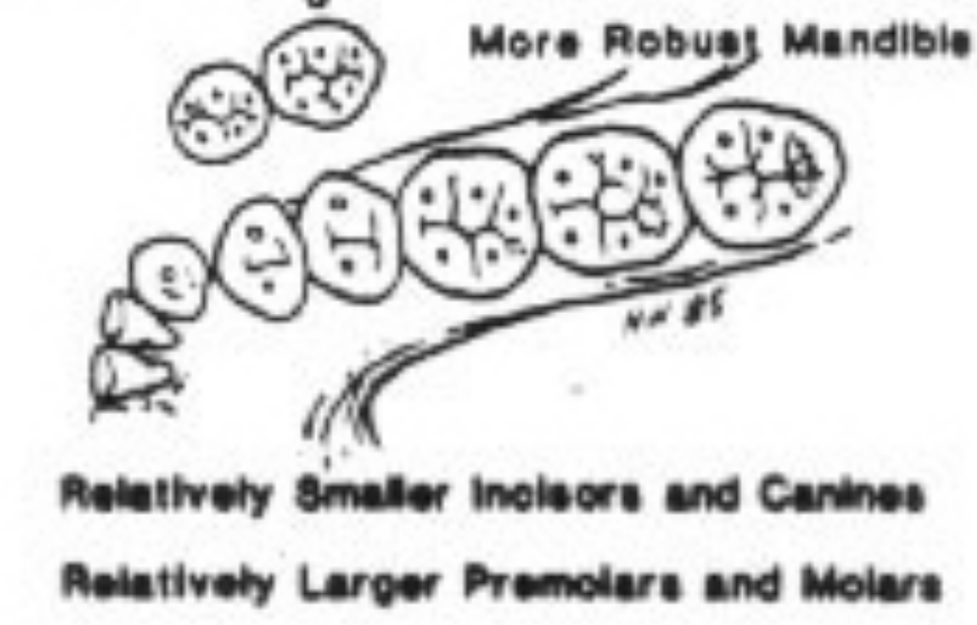
A. robustus









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Australopithecus Vs Paranthropus

Trait	Australopithecus	Paranthropus
Cranial Capacity	450 CC	600 CC
Cranial Vault	High & Expanded	Low & not expanded
Sagital Crest	Present	Absent in Male
Frontal Region	Rounded	Underdeveloped
Supra Orbital Ridge	No Development	Well Developed

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Australopithecus Vs Paranthropus

Trait	Australopithecus	Paranthropus
Jaws	Large	Massive
Simian Shelf	Absent	Present
Dental Arcade	Parabolic	More U-shaped
Foramen Magnum	Hominid like, not yet in centre	Forwardly placed
Foot	Non Divergent Big toe	Divergent big toe

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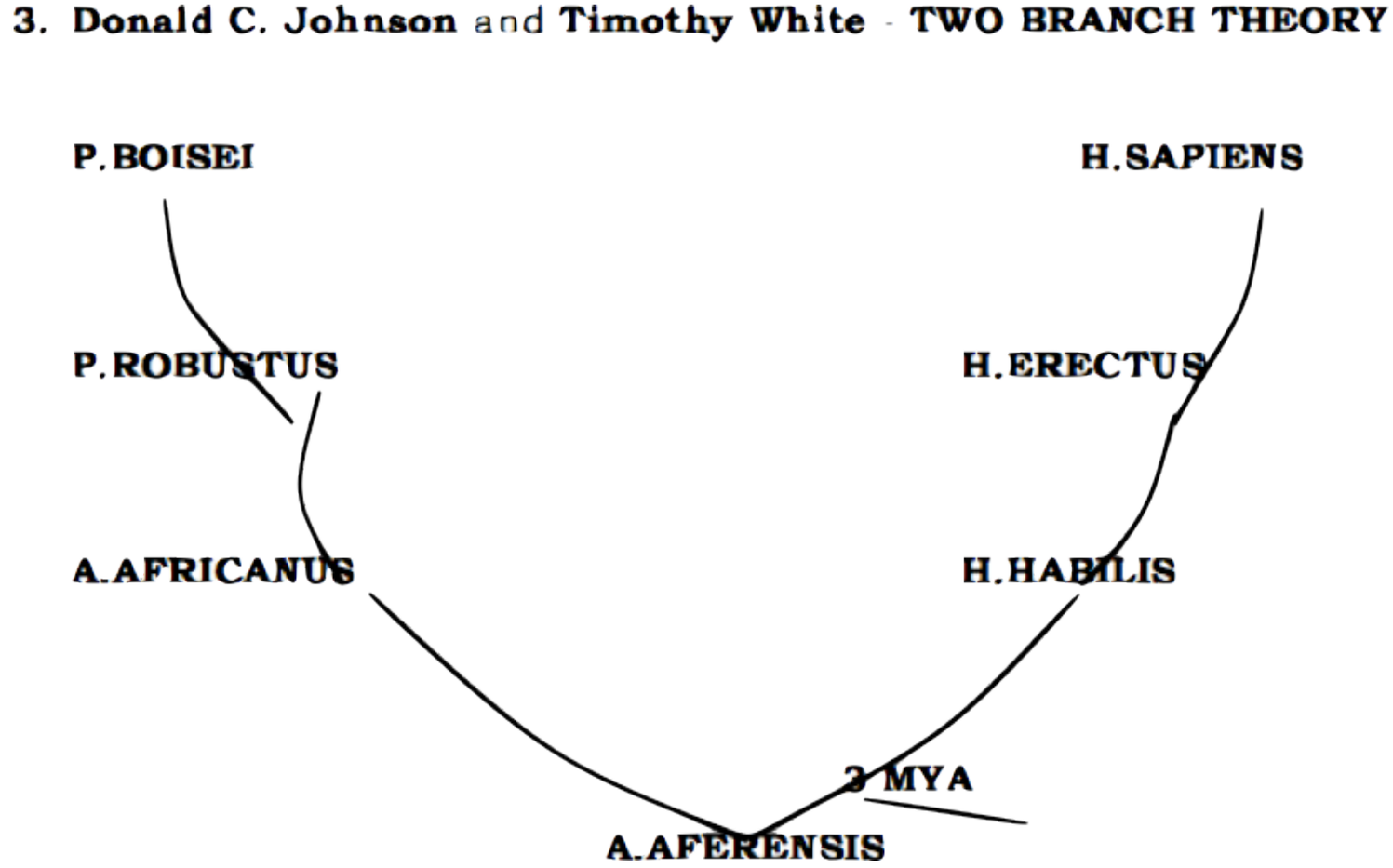
Phylogeny?

- These plio-pliestocene fossils have both hominid and ape like features with overwhelming hominid features
- Presence of ape like features can be accounted for common inheritance from a pongid ancestor



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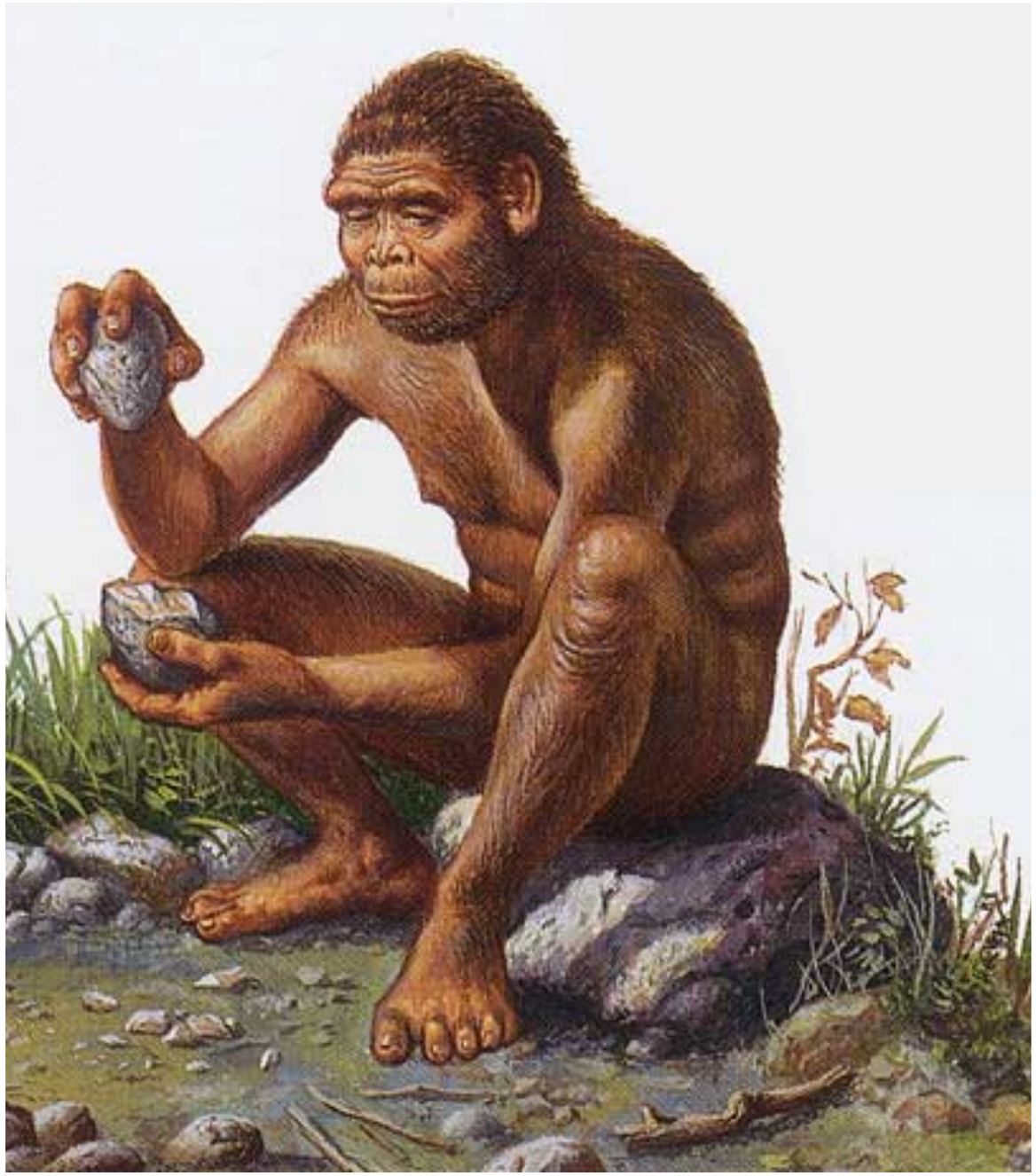
Homo Habilis

- Discovery - 1960 – Leaky's - Olduvai Gorge
- Contemporary of Paranthropus
- **Thought as first real humans**
- Considered direct evolution between Australopithecus and Homo erectus with more primitive paranthropus as evolutionary side line

Climate

- Drier, open environment
- Desert like climate



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Characteristics

- 650-800 CC - cranial capacity
- Well developed frontal lobes (approximating modern man)
- Hominid teeth
- Relatively less massive lower jaws
- Hind limb morphology clearly approached human foot
- Tool maker - Olduwan Tool Industry
- Larger brain - larger gestation - social life?

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Phylogeny

- Great controversy - Homo or Australopithecus
- **Since it has greater cranial capacity than Australopithecus and was a tool user - considered homo**
- Temporal lobes - well developed - centers for memory
- Shows an appealing straight line feel to Australopithecus - Habilis - Erectus evolution
- This period - 2-1.5 MYBP is called crucial humanizing period - What actually happened here? - We need more fossils to clarify the picture or to further confuse it.

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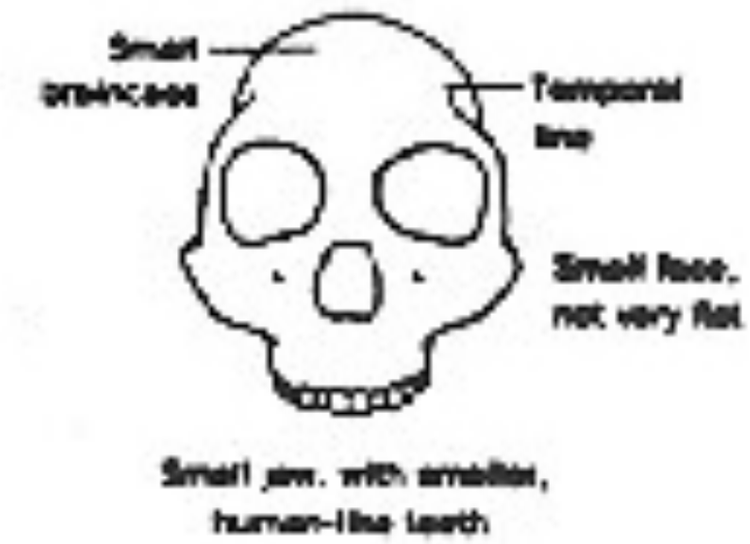
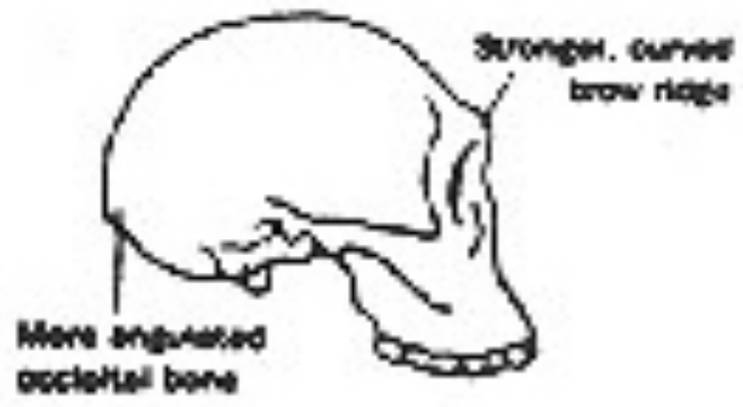
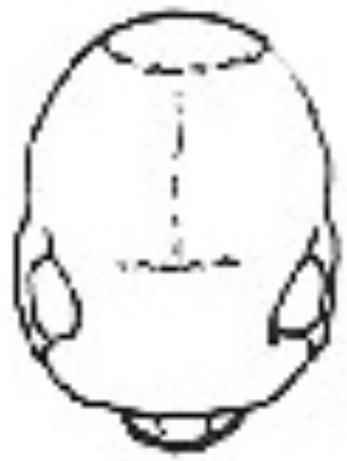
H. habilis







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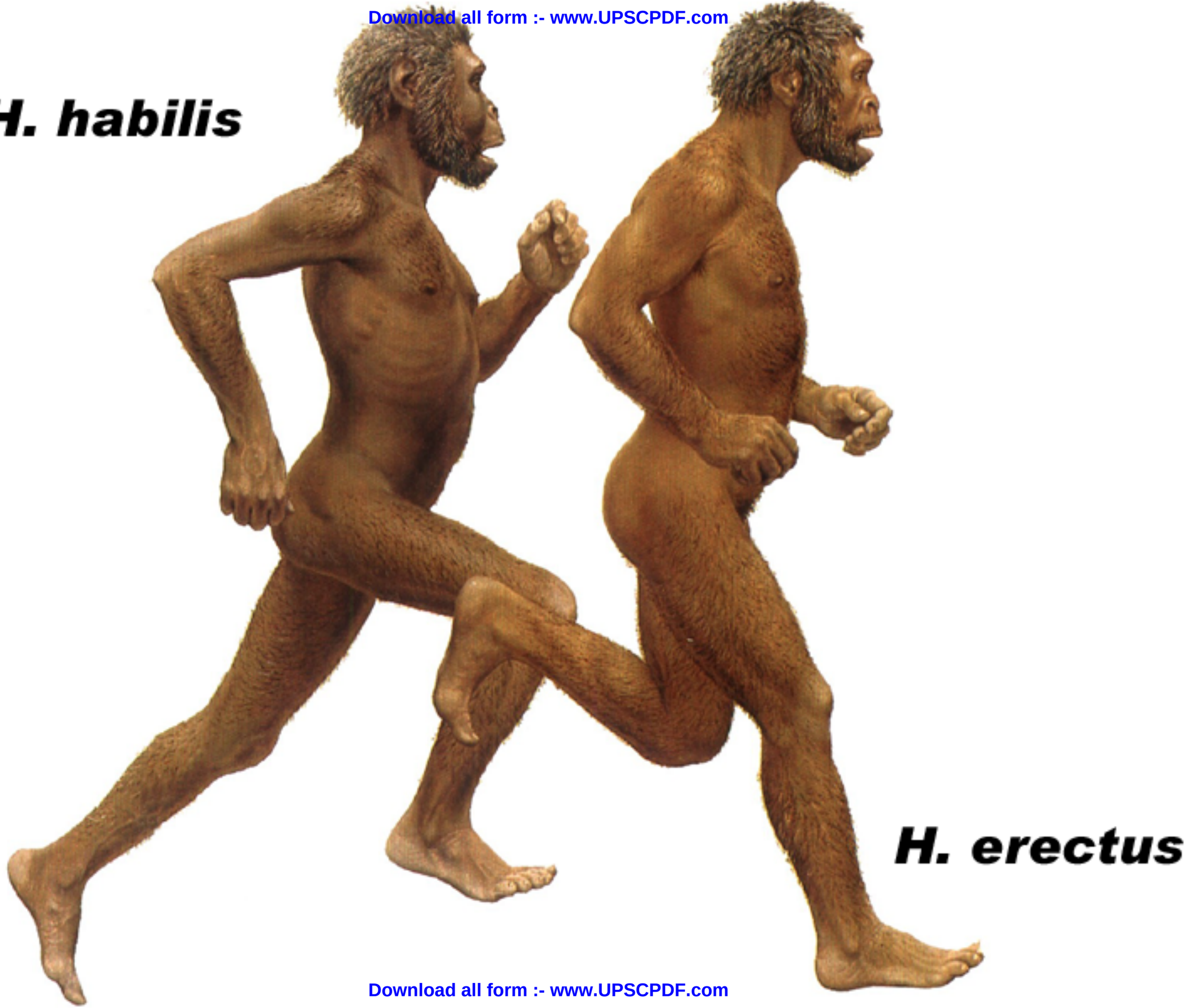

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Homo erectus

- From Java and Peking
- Ranged from Africa to Euro-asia some 1.6 million years to 300,000 years ago
- Middle Pleistocene
- **First formal name of man**
- African specimens are now called **Homo ergaster**

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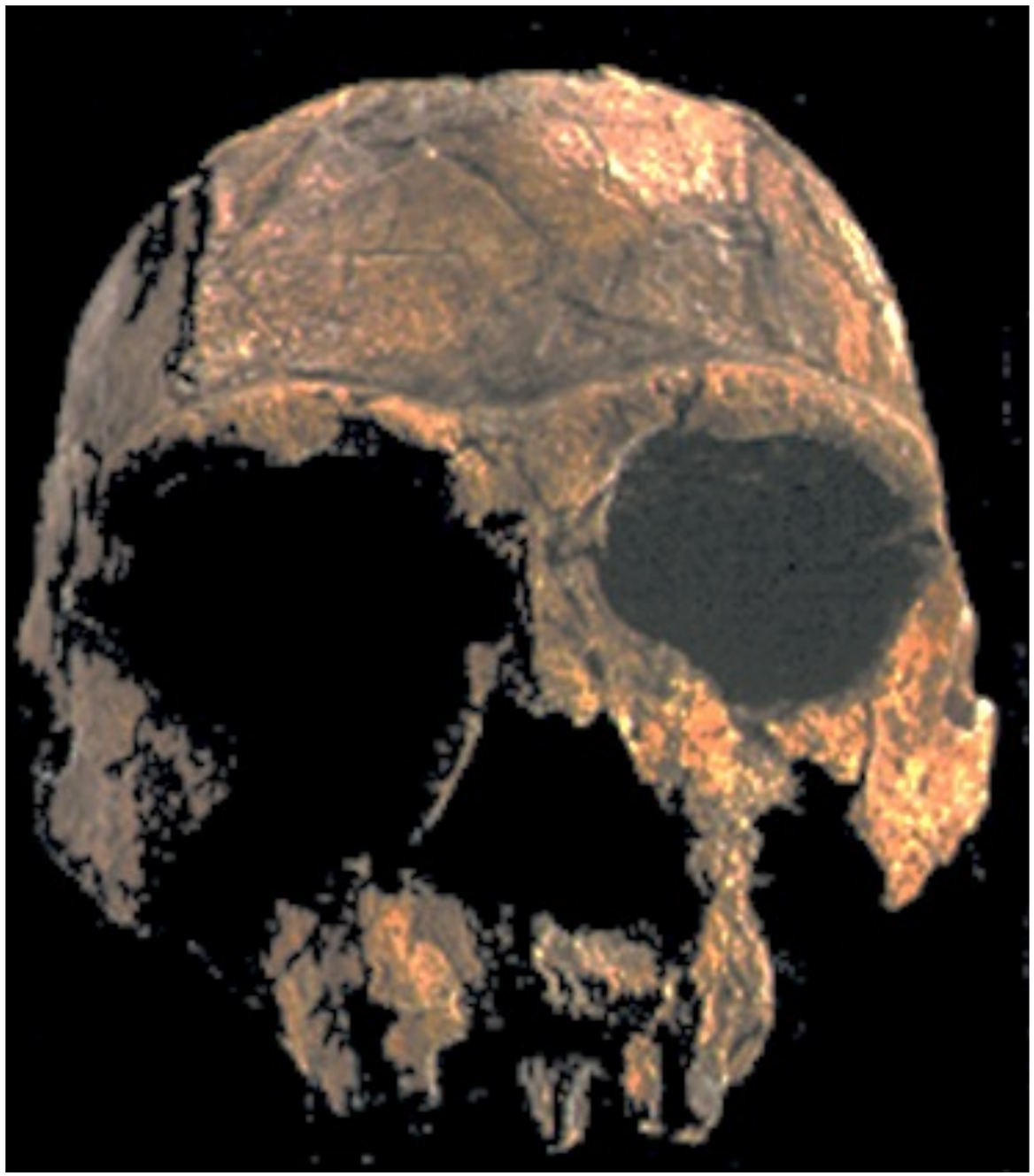
Characterisitcs

- 5 feet tall
- Cranial capacity - 1,000 to 2,000 CC
- Far more adaptable than Australopithecines because existed beyond Africa
- Evolution - both cultural and biological to adapt to new ecological niches
- Peking Man more hominid than Java Man



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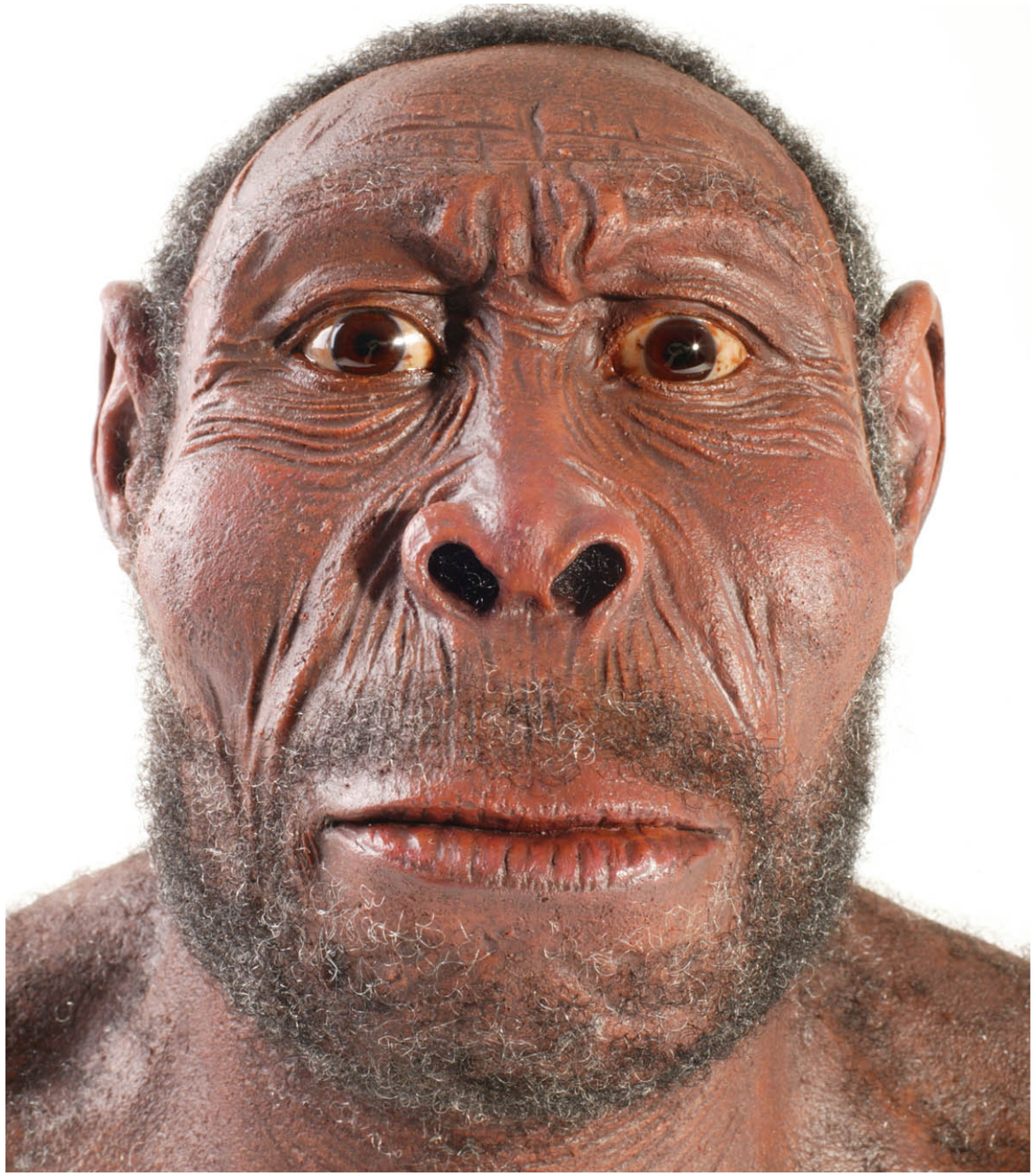
Characteristics

- Beginning of true vertical forehead
- Foramen magnum like modern man - i.e. habitually erect and bipedal
- Parabolic dental arcade
- No diastema in Peking Man
- Most post cranial features - absolutely human
- Vocal communication?
- Big game - group activity
- Religion - burials



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Trait	Java	Peking
Cranial Capacity	775-975 CC	850-1300 CC
Platycephaly	Extreme	Less Pronounced
Sagital Creat	Extreme	Reduced
Supra Orbital	Huge Torus	Reduced
Mandible	Large	Reduced, Modern
Chin	Absent	Beginning
Diastema	Occasional	Absent
Canines	Projecting	Non-projecting

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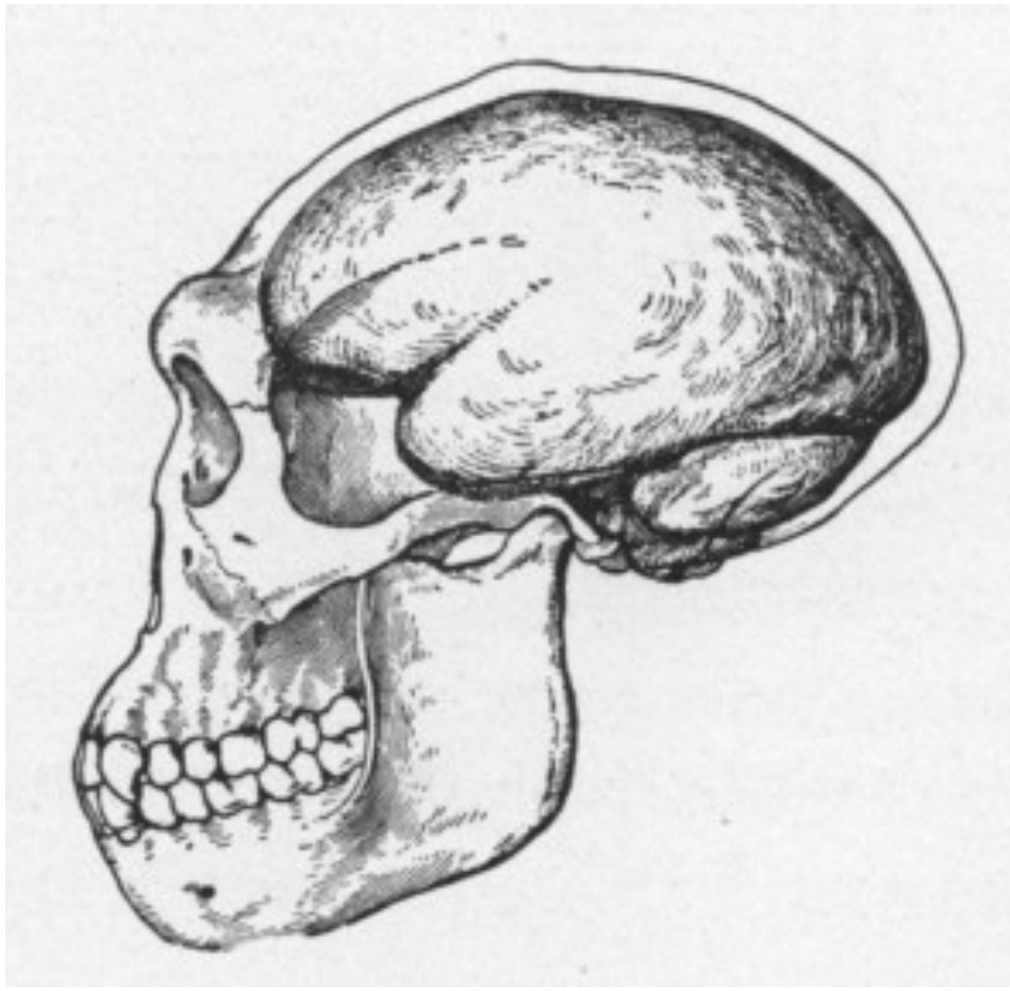
Comparative Skulls

Peking Man

Java Man



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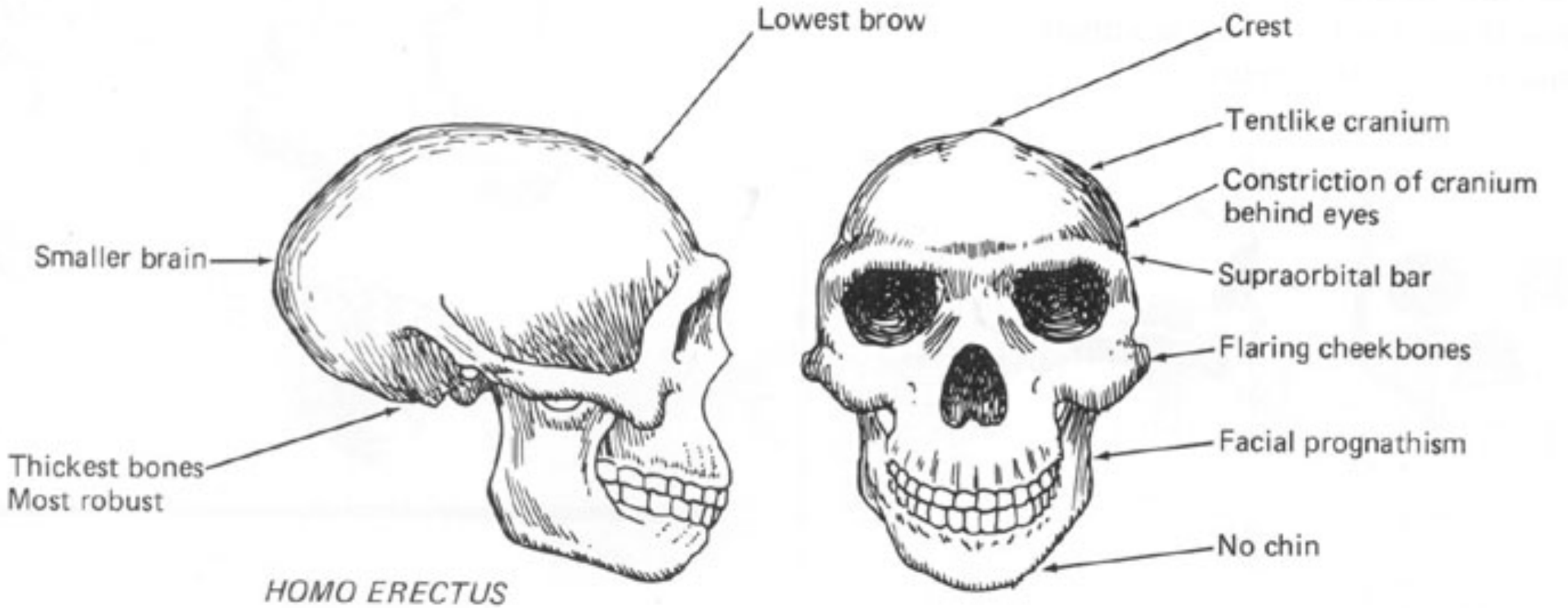

HOMO ERECTUS

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Phylogeny

- No fossils from less than 200,000 years
- What happened to them? and whether they are our ancestors? - hottest area in Paleoanthropology
- One reason why Australopithecines evolved into Homo erectus - Grover Krantz - PERSISTENT HUNTING

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Persistent Hunting

- Persistent hunting requires a commitment to search for hunt - requiring memory of previous successes

- This led to and helped by expanding brain

- Persistent hunting is also supported by bipedalism

- Thus, persistent hunting worked as a selective pressure on expansion of brain and bipedalism

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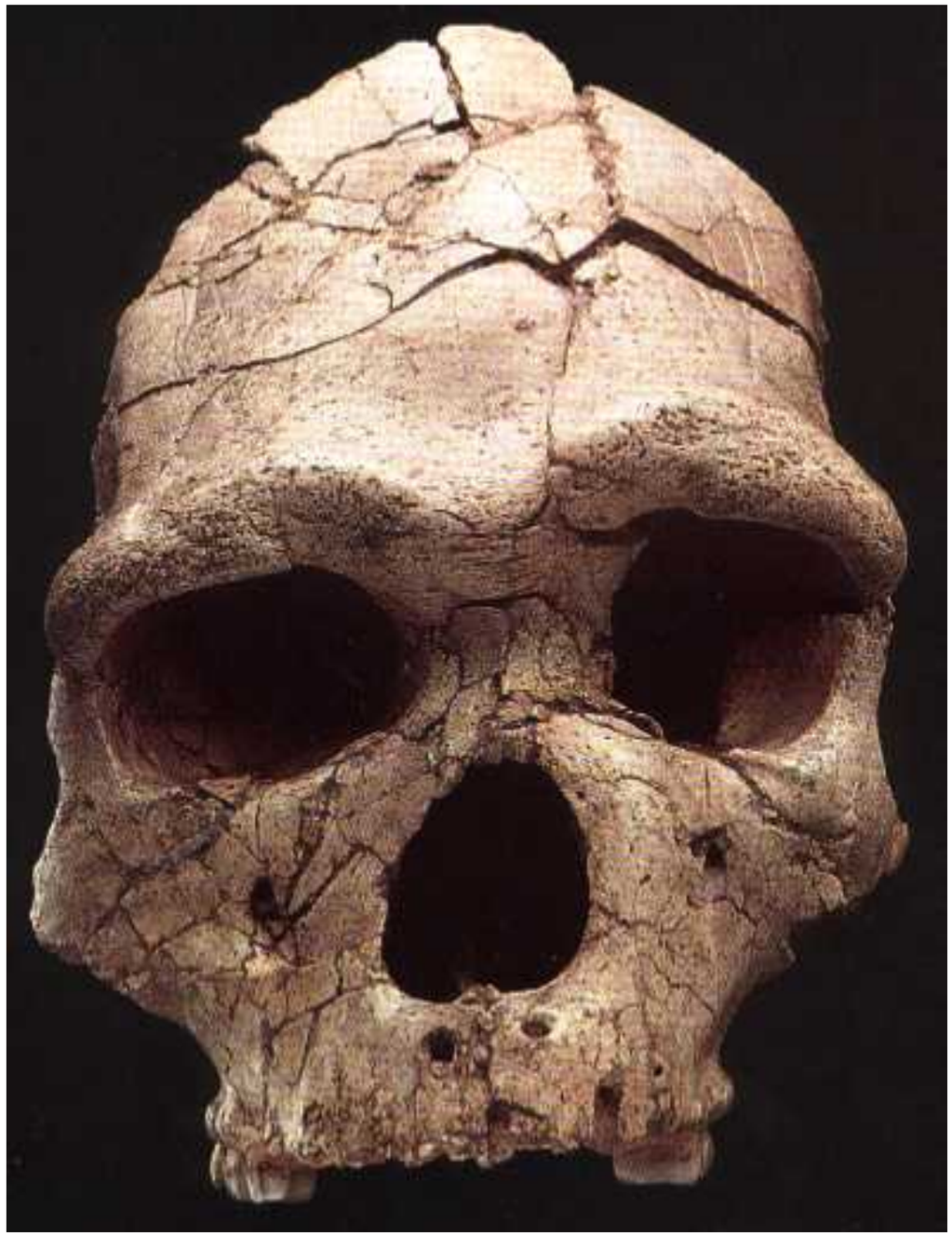
Heidelberg Man

- The fossils of Homo erectus – essentially from Africa and Asia
- Evolution of Homo erectus in Europe – questioned
- 1907 – mandible – from a sand pit – Mauer, near Heidelberg, West Germany
- Dates back to 450,000 years ago



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Characteristics

- Archeological context included elephant, horse, wild boar, bison, deer etc.
- Jaw is massive and general profile is square in shape. Breadth is comparable to modern man.
- Gibbon Like: - Condoloid processes are blunt and rounded and placed at a higher level compared to modern man
- Ape Like: - Truncated angle of jaw with very thick meeting place of two sides of the jaw

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Characteristics

- Human Like...
  - Parabolic dental arcade
  - Absence of diastema
  - No projecting canines
  - Normal size of the teeth

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Phylogeny

- Jaw displays simian features and teeth are hominid
- Opinions vary regarding its phylogeny
- The “Hiedelberg Man” was however a representative of Homo erectus in Europe
- Since it has hominid features, closer to those later perfected in Neanderthals, we can conclude that Heidelberg jaw represents a connecting link between Homo erectus and Neanderthal mosaic in Europe

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The Neanderthals

- The term "Neanderthal", in a strict sense, is used to describe a series of finds from Western Europe dating from the 1,70,000 years ago to the rise of modern man 35,000 years ago

- These have also been called **Classic or Conservative Neanderthals**

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The Neanderthals

- The term Neanderthal, however, has been later extended to include all series of Hominid remains dating about 1,00,000 years to rise of modern man which have been reported outside Western Europe – Asia, Europe and Africa

- The Neanderthals, not conforming to classic type are called **Progressive Neanderthals**

- The Classic Neanderthals are represented best by **La-Chapelle-Aux-Saints** and the Progressive Neanderthals are best represented by **Mount Carmalite Skeletons**

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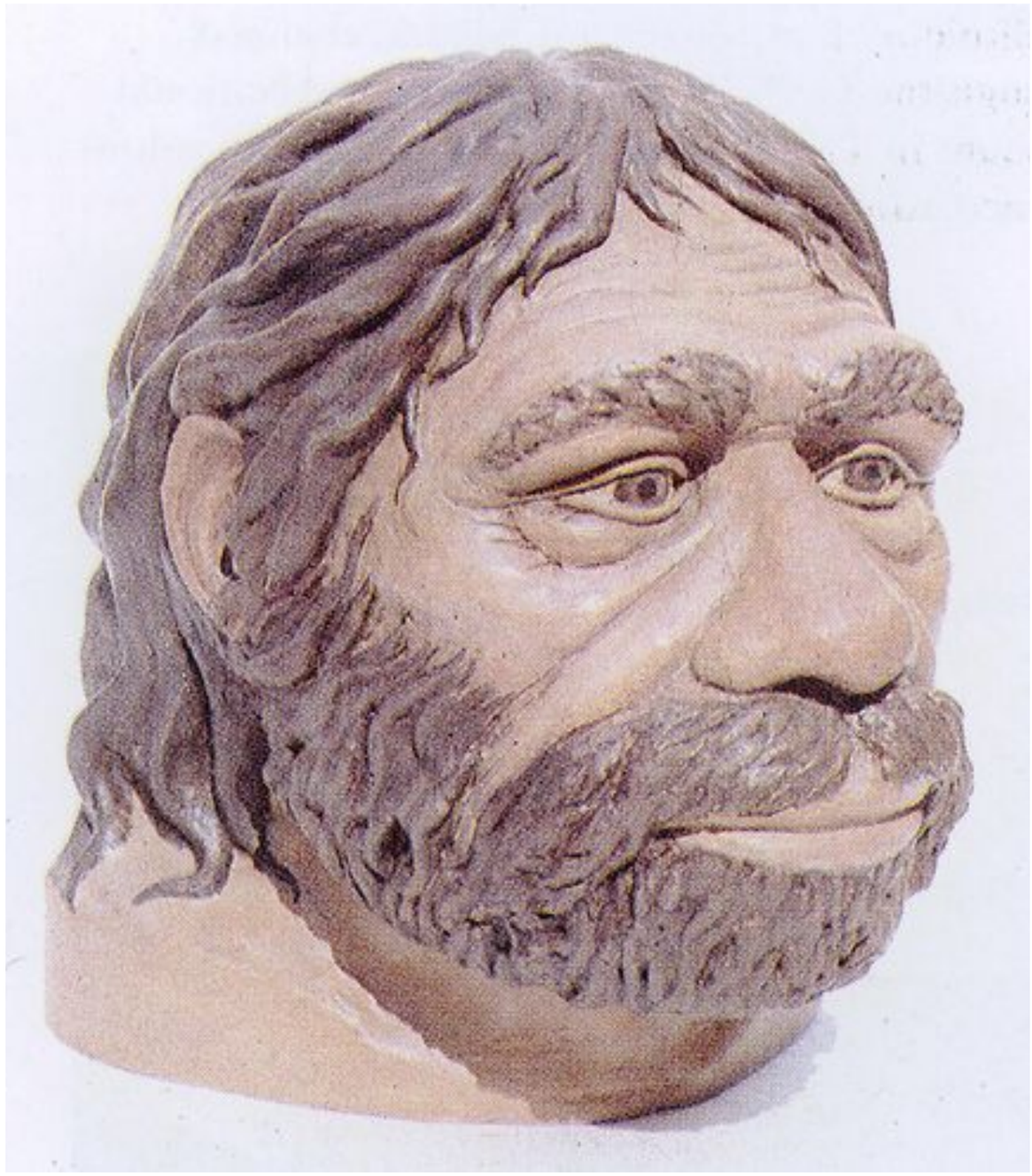
La Chapel Aux Saints

- It's the fossil of an **old man** found in Pogost, 1908, in a small cave of Correze, France
- Archeological context – Mousterian Culture and the fauna included bones of mammals like wooly rhinoceros, reindeer, wild horse and bison



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Characteristics

- Very large skull
- Brain case is elongated and low
- Immense supra-orbital ridges
- Low and retreating fore head
- Protuberant and bun shaped occiput
- Large orbits
- Long and projecting face
- Very broad nose

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Characteristics

- Prognathus upper jaw
- Powerful mandible
- Rudimentary chin
- Foramen magnum is situated farther back in the skull base than in modern man but anterior compared to apes
- Teeth arranged in a broad U, i.e. approaching parabolic arcade
- Condyles are ape like
- Cranial capacity of 1400 CC

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Post Cranial Features

- Spine is short and massive
- Cervical vertebrae have long, almost horizontal, spinous processes similar to chimpanzee – oversized and over balanced jaw
- Thigh bones are strongly bowed forward, as in apes
- Tibiae or shin bones are short and strong – sloped backwards indicating a bent knee walking

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Endocranial Casts

- Left lobe slightly larger – right handedness
- Frontal lobes less expanded compared to others but its areas of speech is well developed
- Temporal lobe greatly expanded indicating greater auditory eminence
- Visual and visuo-psychic areas are increased
- Though brain is primitive, it is distinctively human

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Phylogeny

- This interpretation precludes a direct evolutionary connection between Neanderthal and modern humans
- Blow to Boule's interpretation – 1957 – Strauss and Cave reanalyzed La Chappele Aux Saints and they concluded that the skeleton was that of a robust male 40-50 years old who had suffered from severe and extensive arthritis!!!

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Progressive Neanderthals

- Neanderthal material outside Western Europe has usually been referred to as Progressive because it is less extreme than the Conservative type and exhibits many features of modern man
- Found buried in adjoining caves in Jerusalem
- The first cave SKUHL yielded fragmentary remains of ten individuals
- The second cave TIBUN yielded skeleton of an adult female, mandible of an adult male, some bones and teeth
- The Skuhl material represented Progressive Neanderthals while the Tibun represented Conservative type
- The stone tool industries were same in both caves

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Characteristics

- More taller than conservatives
- Relatively massive heads, and larger face
- Prominent and vertical foreheads
- Very large eyebrow ridges
- Face is of moderate length and is orthognathus
- Orbits are placed low compared to classic type

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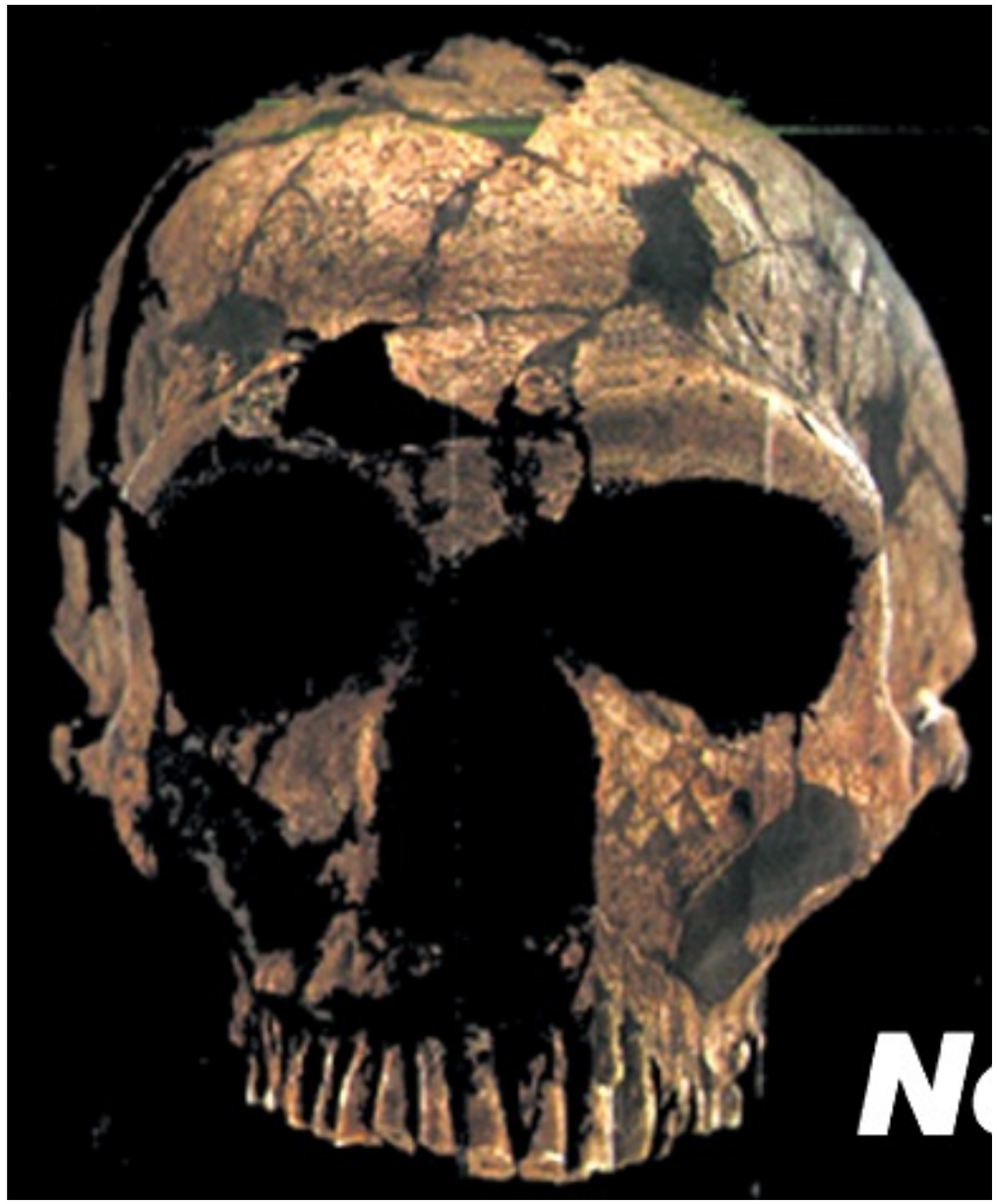
Characteristics

- Jaws of moderate size
- Prominent chin
- Cranial capacity – 1600 CC
- Endocranial casts suggest that brain lacks primitive simian features found in conservative type
- Limb structures are predominantly like that of modern man



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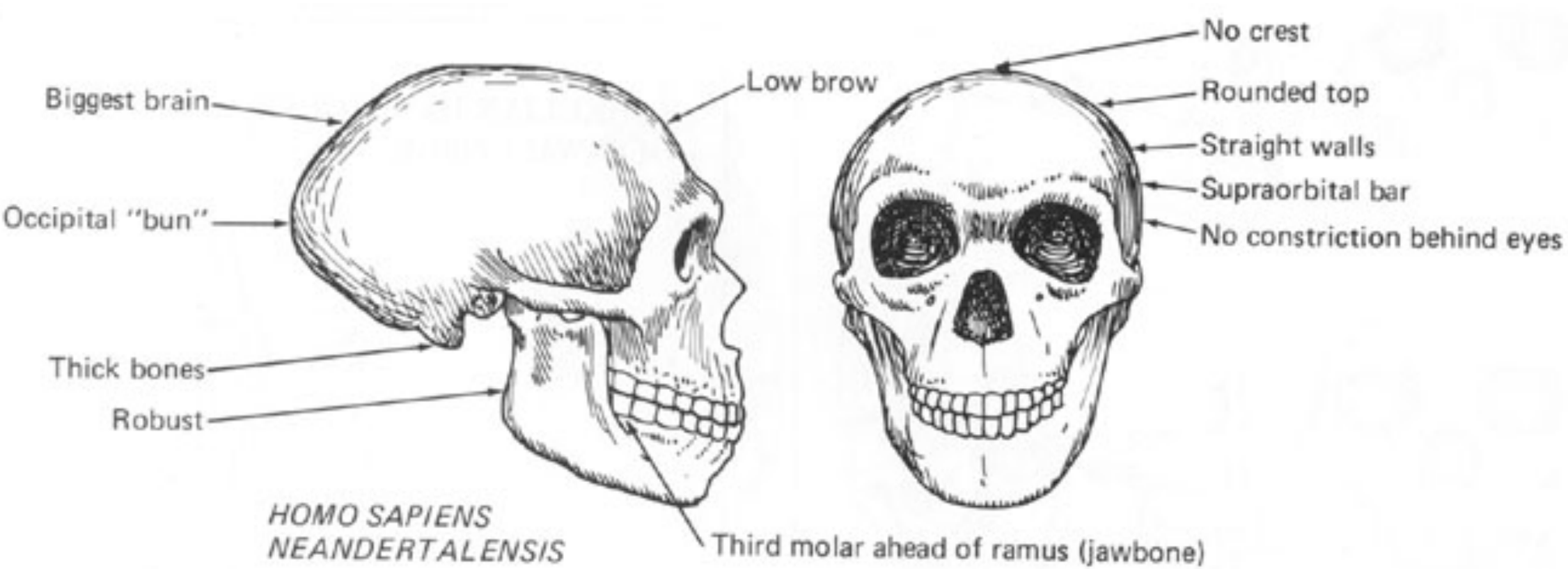

HOMO SAPIENS NEANDERTALENSIS

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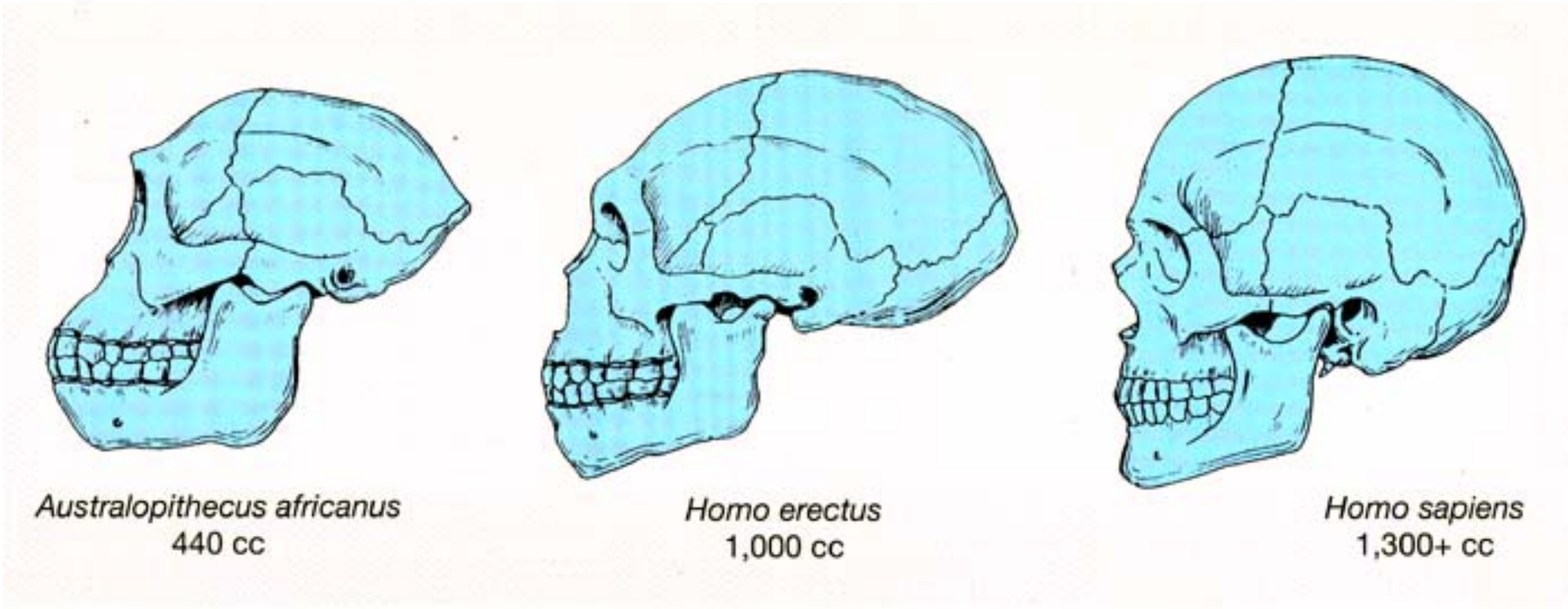

Australopithecus africanus
440 cc

Homo erectus
1,000 cc

Homo sapiens
1,300+ cc

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Disappearance of Neanderthals

- The Neanderthals seem to have disappeared between 35,000 to 40,000 years ago as we have no fossil evidence thereafter
- According to one view, they are believed to have been driven out by the migrations of advanced modern forms which are now referred to as Homo sapiens sapiens
- Some others believe that Neanderthals gave rise to modern man

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Neanderthal Phylogeny

Neanderthal Phase of Man Hypothesis

- Schwalbe - 1904
- This is a unilinear view
- Neanderthals arose from middle Pleistocene predecessors by successive evolution
- They passed through the Neanderthal phase and finally evolved into the modern man

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Neanderthal Phylogeny

Pre-Neanderthal Hypothesis

- Sergi, Howell and Le Gros Clark
- The Neanderthals arose from a pre-Neanderthal stock that progressively specialized towards cold resistance and was open to severe natural selection
- The restricted gene flow resulted into a Classic Neanderthal isolate represented by La Chappelle Aux Saints
- This specialized Neanderthal offshoot represents a group sharing new traits of sub-specific taxonomic value
- The parent line may have developed outside Europe - Africa
- The pre Neanderthal stock is early Homo sapiens from Europe

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Neanderthal Phylogeny

Pre-Sapien Hypothesis

- Boule, Vallois, Weiner
- European modern sapient lineage represented by Swanscombe and Steinheim existed separately from the Neanderthals and ultimately gave rise to modern Europeans
- Neanderthals, however, became extinct
- However, it was later found that Swanscombe and Steinheim had Neanderthal features and were not anatomically fully modern

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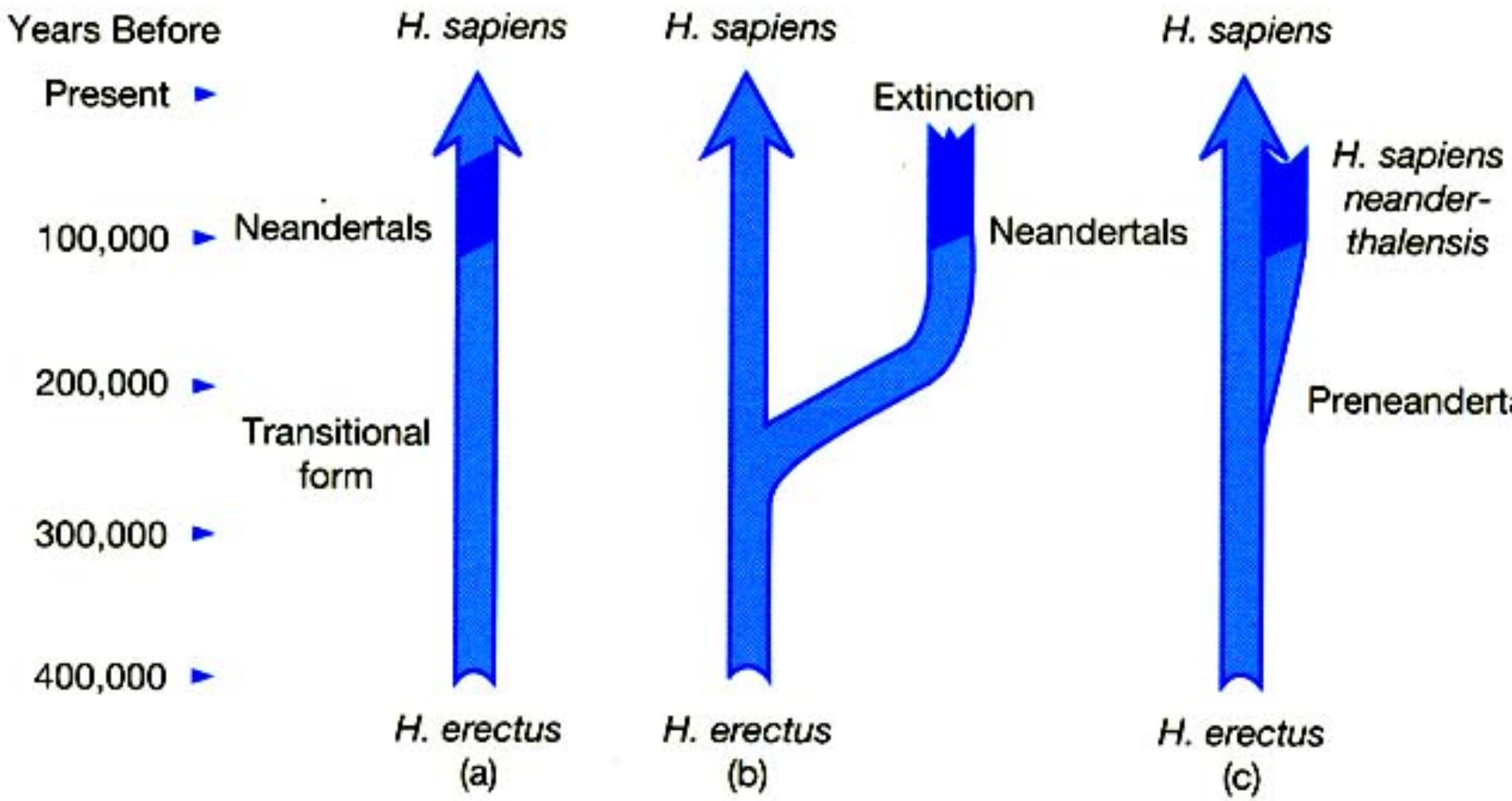
Neanderthal Phylogeny



Evolution of Neandertals and Humans

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Homo erectus
KNM-ER-3733
Koobi Fora, Kenya

1.8 Ma – 800 ka

Homo heidelbergensis
Petralona
Greece

800 – 130 ka

Homo neanderthalensis
Amud
Israel

130 - 30 ka

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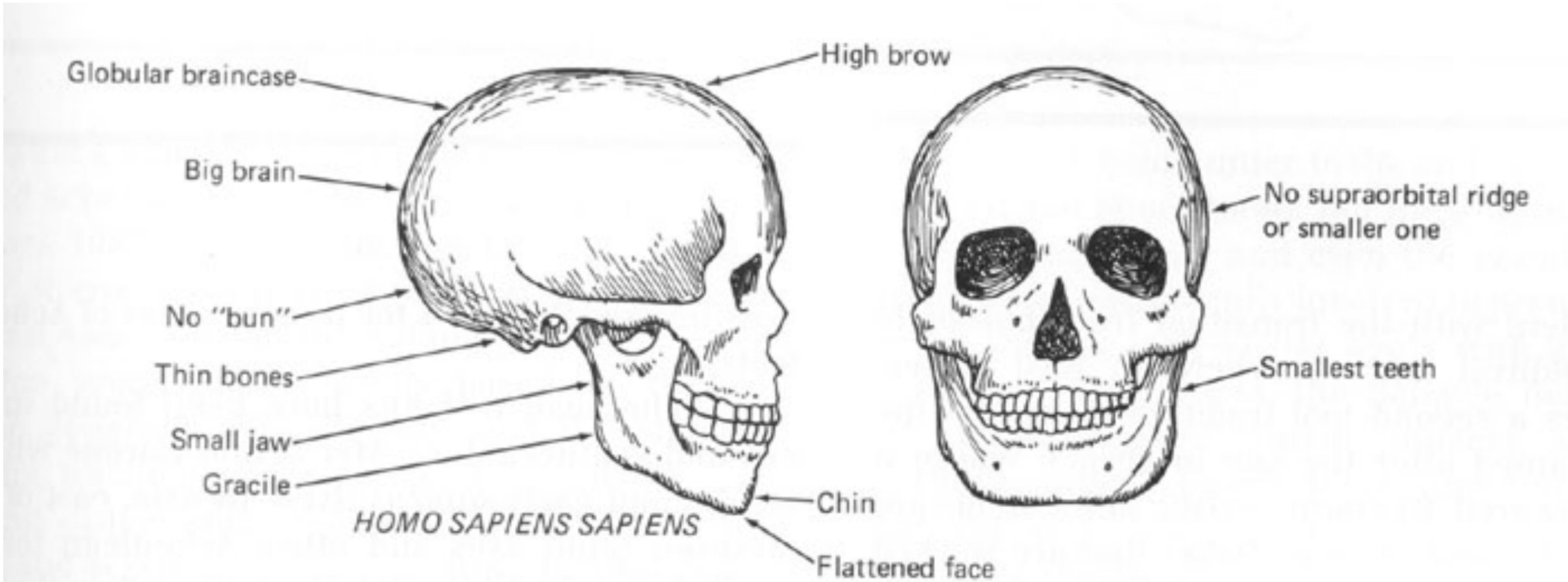

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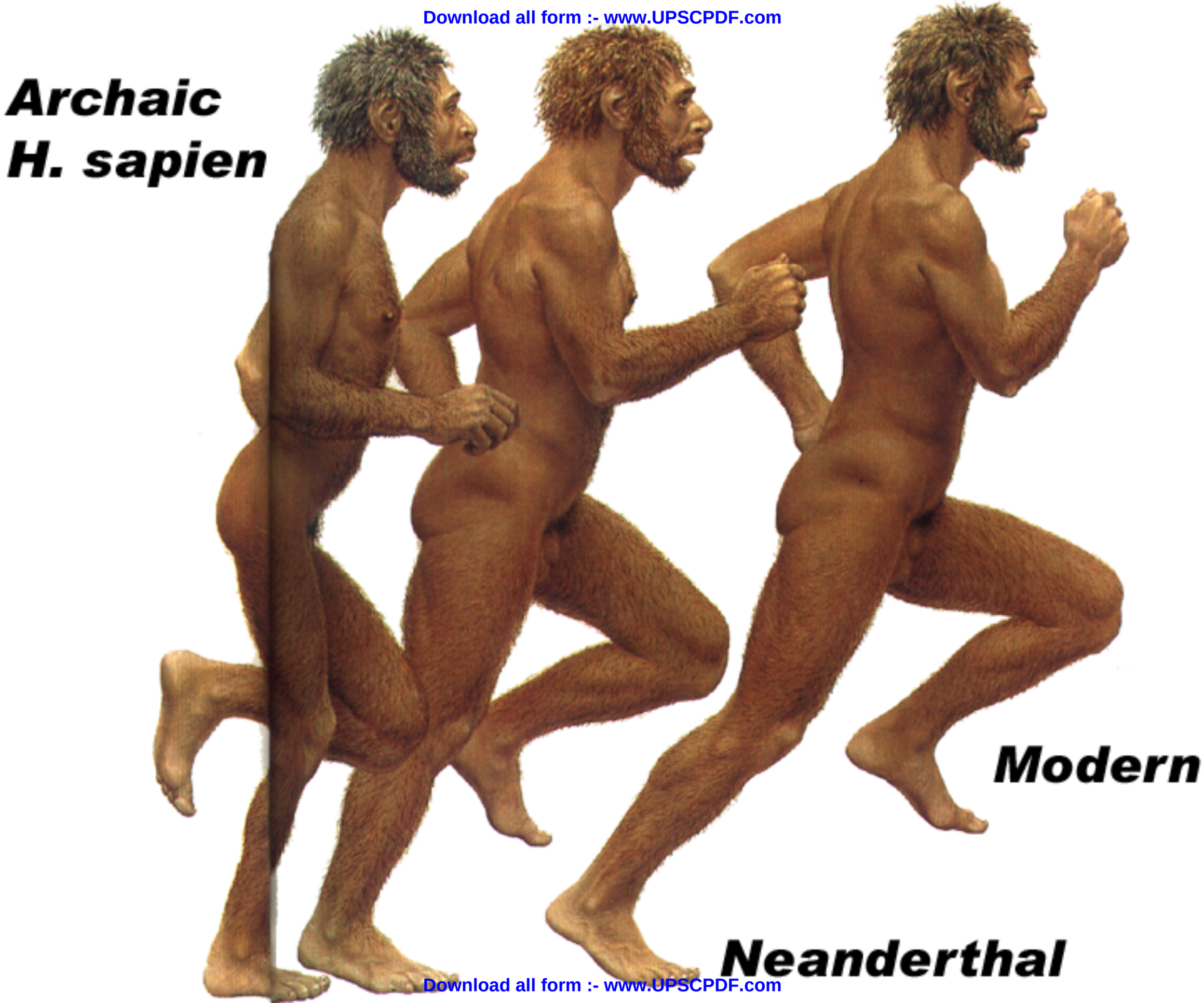

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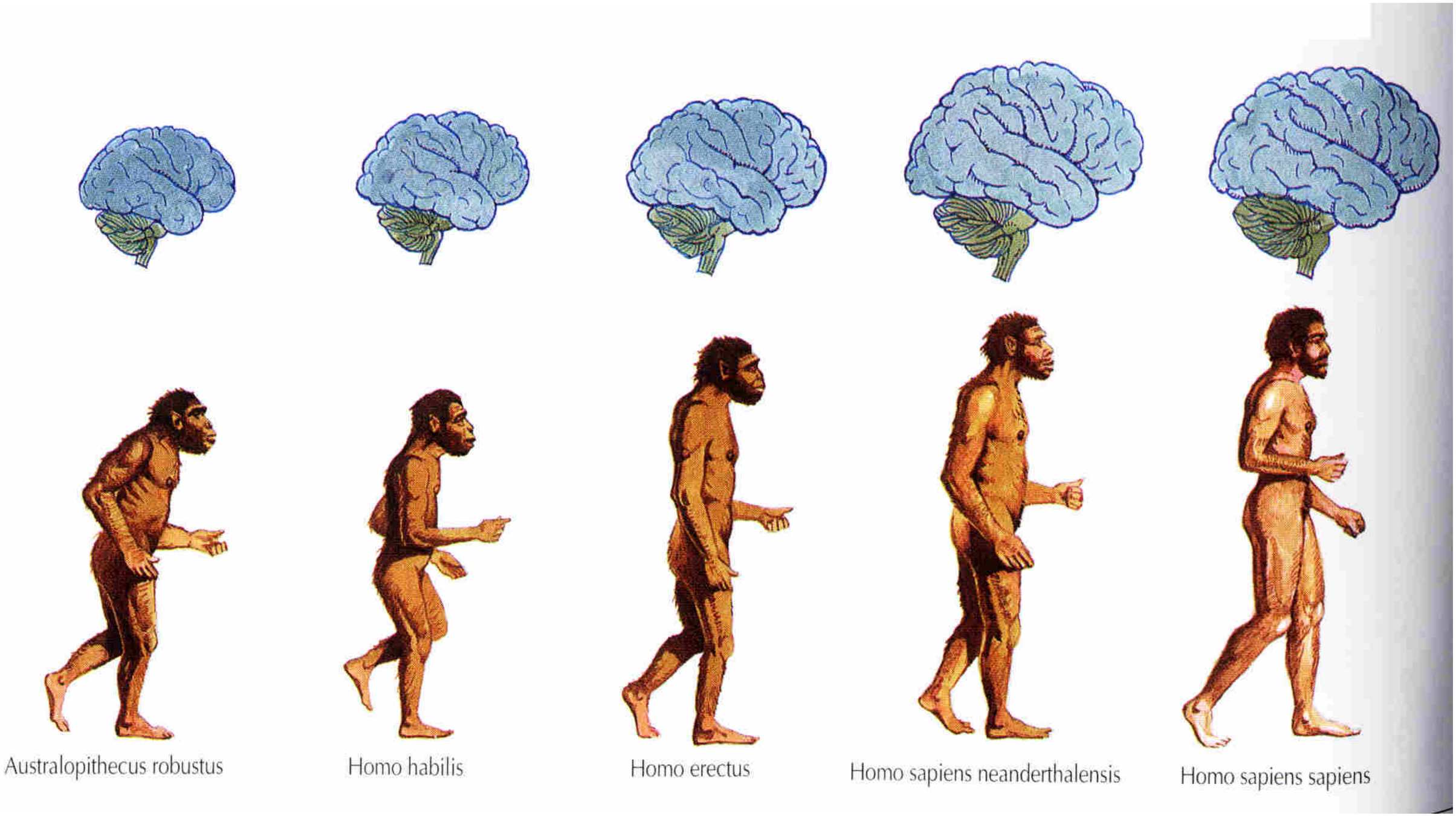

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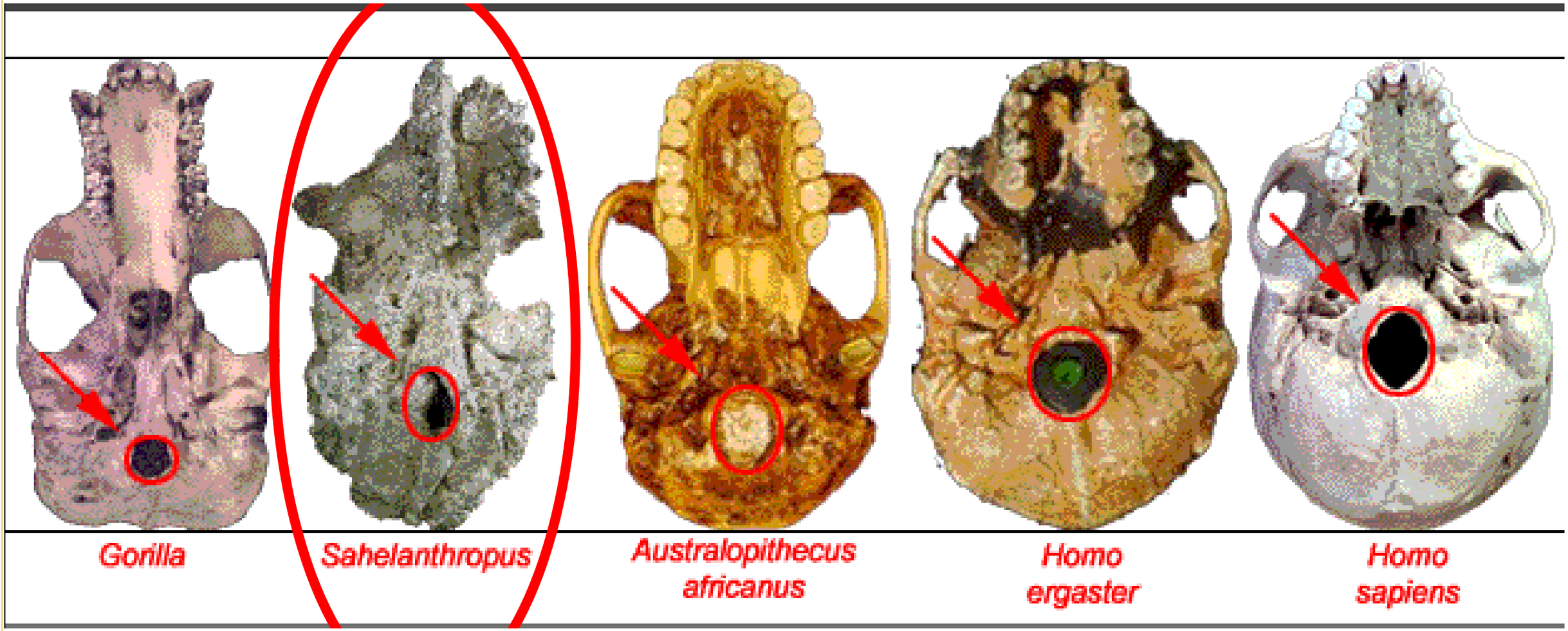
Position of Foramen Magnum



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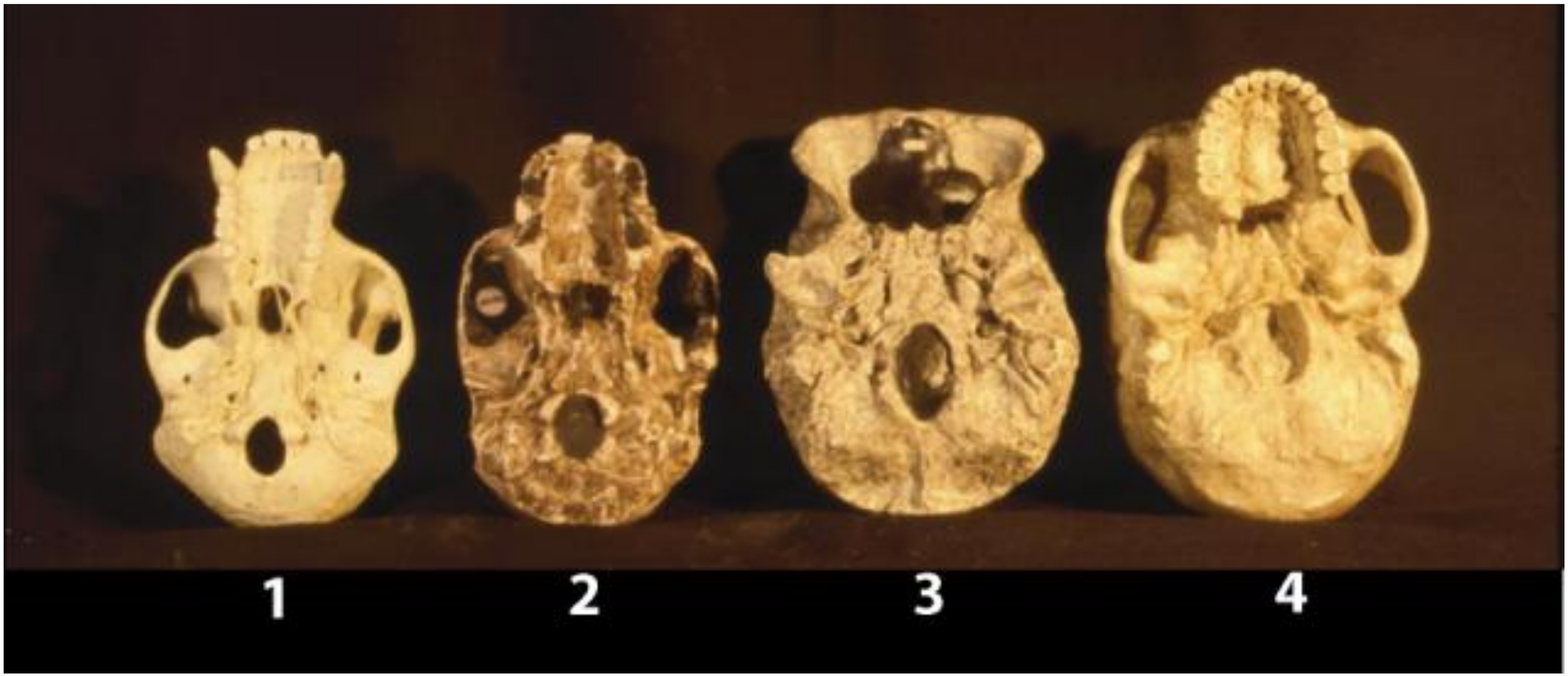

1. Chimp; 2. Australopithecus; 3. H. erectus; 4. H. sapiens

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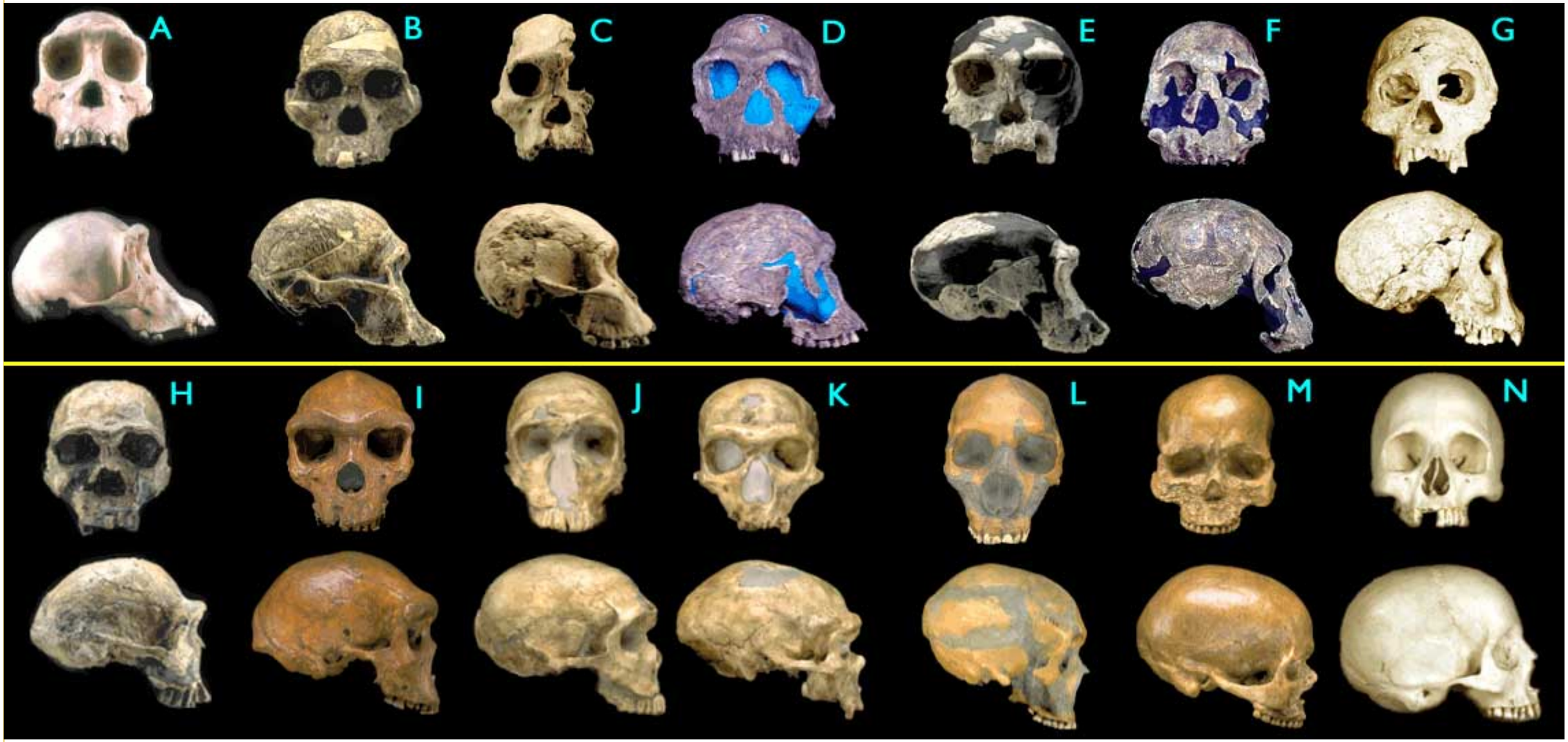

(A) Pan troglodytes, chimpanzee, modern
(B) Australopithecus africanus, STS 5, 2.6 My
(C) Australopithecus africanus, STS 71, 2.5 My
(D) Homo habilis, KNM-ER 1813, 1.9 My
(E) Homo habilis, OH24, 1.8 My
(F) Homo rudolfensis, KNM-ER 1470, 1.8 My
(G) Homo erectus, Dmanisi cranium D2700, 1.75 My
(H) Homo ergaster (early H. erectus), KNM-ER 3733, 1.75 My
(I) Homo heidelbergensis, "Rhodesia man," 300,000 - 125,000 y
(J) Homo sapiens neanderthalensis, La Ferrassie 1, 70,000 y
(K) Homo sapiens neanderthalensis, La Chappelle-aux-Saints, 60,000 y
(L) Homo sapiens neanderthalensis, Le Moustier, 45,000 y
(M) Homo sapiens sapiens, Cro-Magnon I, 30,000 y
(N) Homo sapiens sapiens, modern

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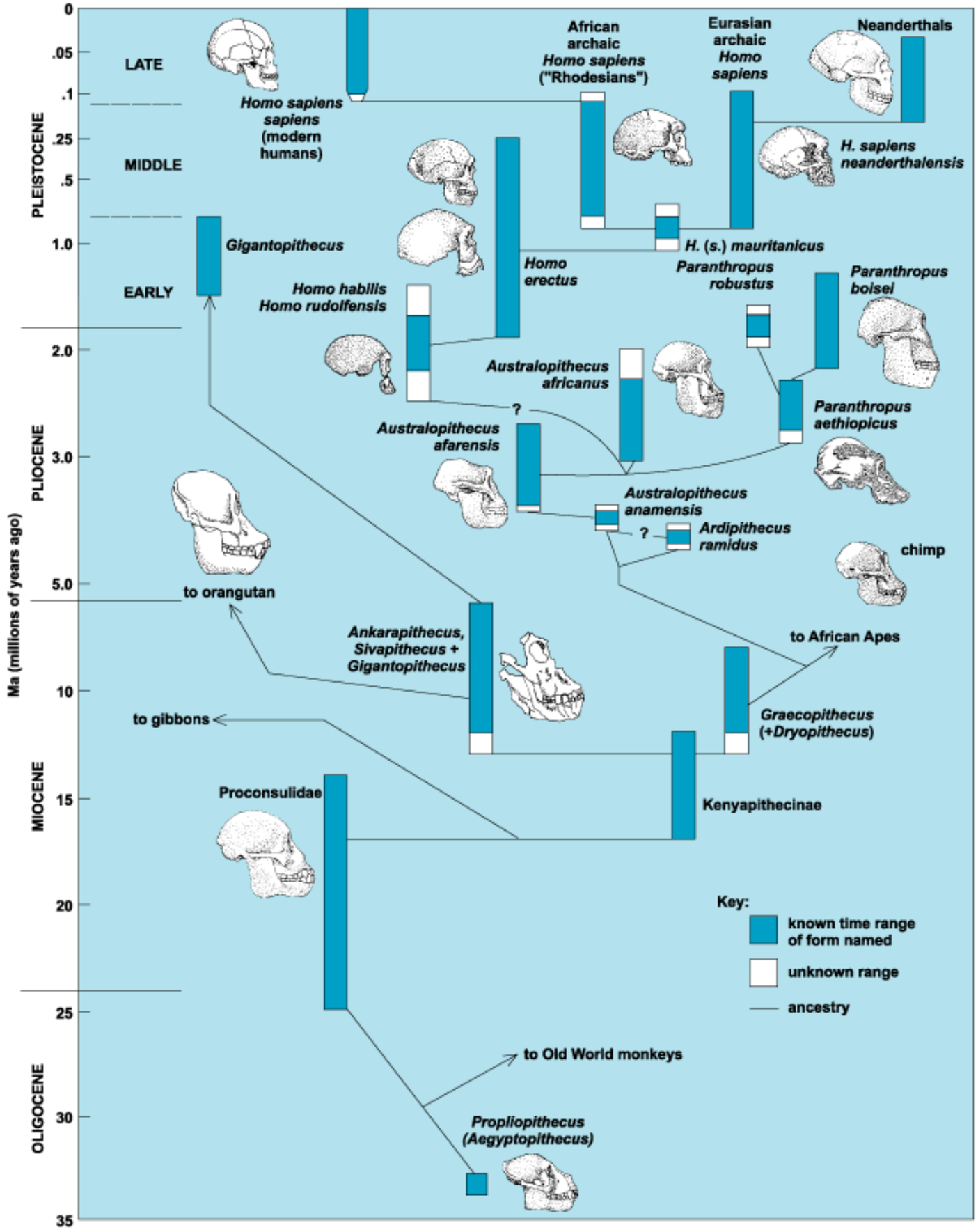

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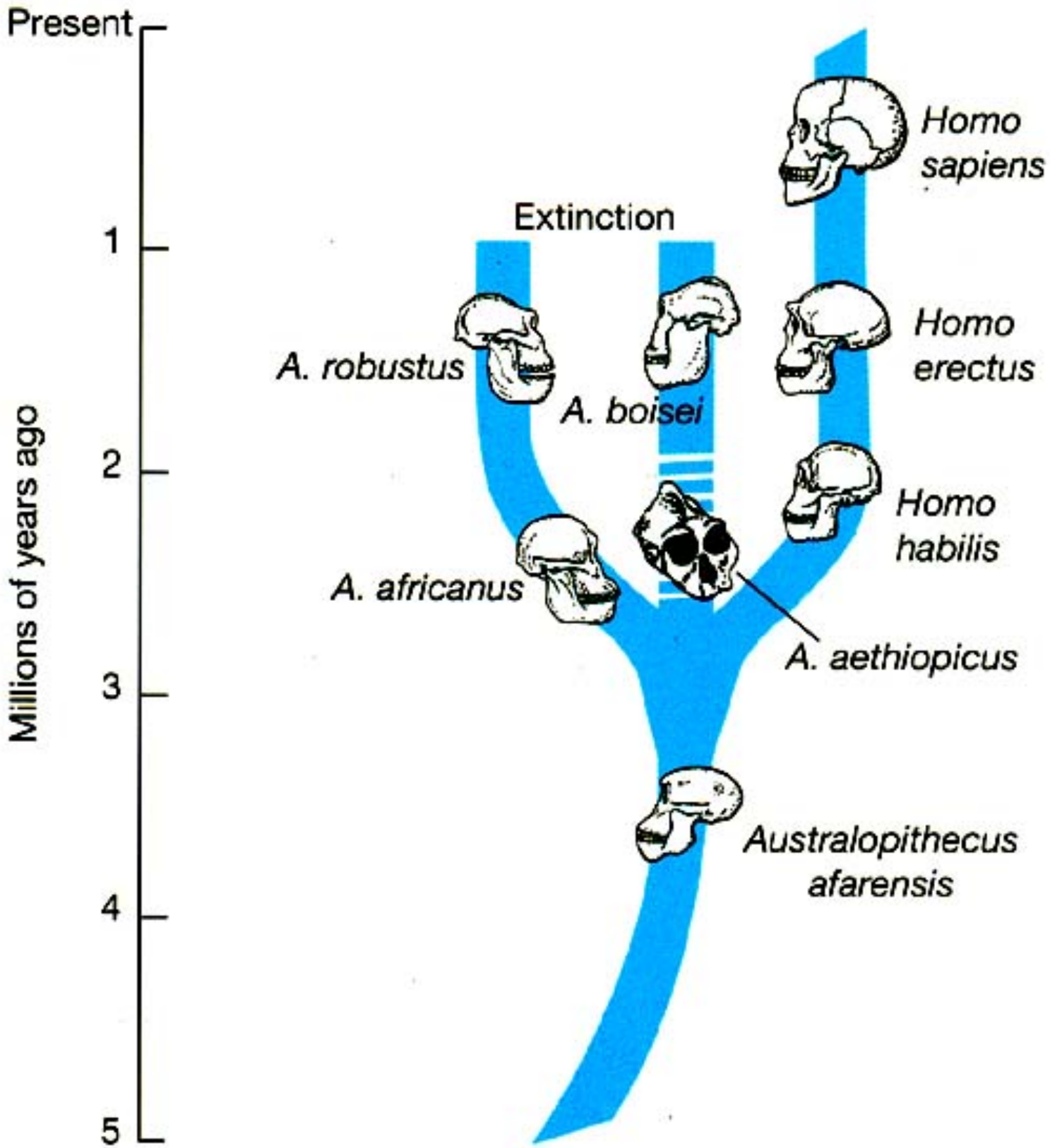

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Species	Date	Body size	Brain size	Tools
Pan troglodytes				
Ardipithecus ramidus	4.4-4.0 Ma	40 kg.	? (ape-like)	None
Australopithecus anamensis	3.9-3.1 Ma	45-60 kg.	? (ape-like)	?
Australopithecus afarensis	3.5-3.0 Ma	25-60 kg.	400 - 500 cc.	None
Australopithecus africanus	2.8-2.3 Ma.	25-40 kg.	? 450 cc.	None
Australopithecus robustus	1.8-1.6 Ma.	30-40 kg.	550 - 600 cc.	Oldowan
Australopithecus boisei	1.8-1.0 Ma.	50-70 kg.	400 - 500 cc.	Oldowan
Australopithecus aethiopicus	2.6-2.4 Ma.	50-70 kg.	400 cc.	None
Homo habilis	1.9-1.6 Ma.	40-50 kg.	> 600 cc.	Oldowan.
Homo rudolfensis	1.9 Ma.	50 kg.	750 cc.	Oldowan?
Homo ergaster	1.9-1.2 Ma.	large	> 850 cc.	Acheulian?
Homo erectus	1.8-0.8 Ma.	~ human	900 - 1200 cc.	choppers and scrapers
Archaic Homo sapiens	0.5-0.1 Ma.	human	1200 - 1500 cc.	Acheulian and Mousterian
Homo neanderthalensis	0.3-0.03 Ma.	human	1200 - 1600 cc.	Mousterian
Homo sapiens sapiens	0.1-0.0 Ma.	human	1300 - 1500 cc.	Mousterian and later

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